Smooth muscle and the cardiovascular and lymphatic systems

CHAPTER 6

The cardiovascular system carries blood from the heart to all parts of the body through a series of tubes, all but the smallest of which are muscular. The muscle in these tubes is of two types: smooth muscle is characteristic of the walls of blood vessels, whereas cardiac muscle provides the walls of the heart chambers with their powerful contractile pumping action. The general characteristics and classification of muscle tissues are given on page 103. Smooth muscle also forms an important contractile element in the walls of many other organ systems of the body, e.g. the gastrointestinal tract and the airways.

SMOOTH MUSCLE

In smooth muscle tissue, the contractile proteins actin and myosin are not organized into regular sarcomeres, visible as transverse striations, and so the cytoplasm has a smooth (unstriated) appearance. Smooth muscle is also referred to as involuntary muscle because its activity is neither initiated nor monitored consciously. It is much more variable, in both form and function, than either striated or cardiac muscle, a reflection of its varied roles in different systems of the body.

Smooth muscle is typically found in the walls of tubular structures and hollow viscera. It regulates diameter (e.g. in blood vessels, and branches of the bronchial tree); propels liquids or solids (e.g. in the ureter, hepatic duct and intestines); or expels the contents (e.g. in the urinary bladder and uterus). The actual arrangement of the cells varies with the tissue. The account that follows will therefore be concerned with the generic properties of smooth muscle. The more specialized morphologies of smooth muscle are described in the appropriate regional chapters.

MICROSTRUCTURE OF SMOOTH MUSCLE

Smooth muscle cells (fibres) are smaller than those of striated muscle. Their length can range from 15 µm in small blood vessels to 200 µm, and even to 500 µm or more in the uterus during pregnancy. The cells are spindle-shaped, tapering towards the ends from a central diameter of 3–8 µm (Fig. 6.1). The nucleus is single, located at the midpoint, and often twisted into a corkscrew shape by the contraction of the cell. Smooth muscle cells align with their long axes parallel and staggered longitudinally, so that the wide central portion of one cell lies next to the tapered end of another. Such an arrangement achieves both close packing and a more efficient transfer of force from cell to cell. In transverse section, smooth muscle is seen as an array of circular or slightly polygonal profiles of very varied size, and nuclei are present only in the centres of the largest profiles (Fig. 6.2). This appearance contrasts markedly with that of skeletal muscle cells, which show a consistent diameter in cross-section and peripherally placed nuclei throughout their length.

Smooth muscle has no attachment structures equivalent to the fasciae, tendons and aponeuroses associated with skeletal muscle. There is a special arrangement for transmitting force from cell to cell and, where necessary, to other soft tissue structures. Cells are separated by a gap of 40–80 nm. Each cell is covered almost entirely by a prominent basal lamina, which merges with a reticular layer consisting of a network of fine elastin, reticular fibres (collagen type III) and type I collagen fibres (Fig. 6.3). These elements bridge the gaps between adjacent cells and provide mechanical continuity throughout the fascicle. The cell attaches to components of this extracellular matrix at dense plaques (**Figs 6.3A and 6.4**), where the basal lamina is thickened; cell–cell attachment occurs at intermediate junctions or desmosomes, formed of two adjacent dense plaques. At the boundaries of fascicles, the connective tissue fibres become interwoven with those of interfascicular septa, so that the contraction of different fascicles is communicated throughout the tissue and to neighbouring structures. The components of the reticular network, the ground substance and collagen and elastic fibres, are synthesized by the smooth muscle cells themselves, not by fibroblasts or other connective tissue cells, which are rarely found within fasciculi.

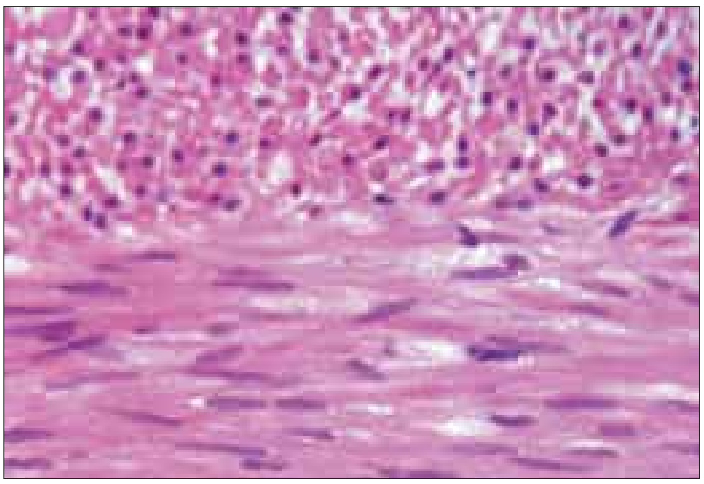

Fig. 6.1 Smooth muscle cells (fibres) in longitudinal (bottom) and transverse (top) section at the boundary of circular and longitudinal muscle layers in the human intestinal wall. Individual cells are spindle-shaped with a single central nucleus, aligned in parallel with neighbouring cells in a fasciculus. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

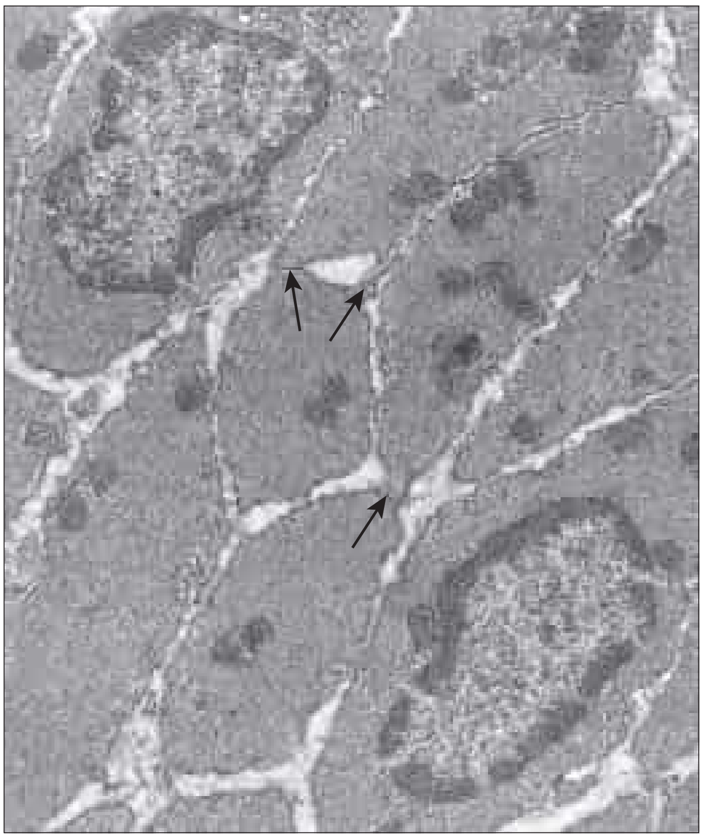

Fig. 6.2 A transmission electron micrograph showing smooth muscle fibres in transverse section, two at the level of their single central nucleus. In several places, the plasma membranes of adjacent cells are closely approximated at gap junctions (arrows).

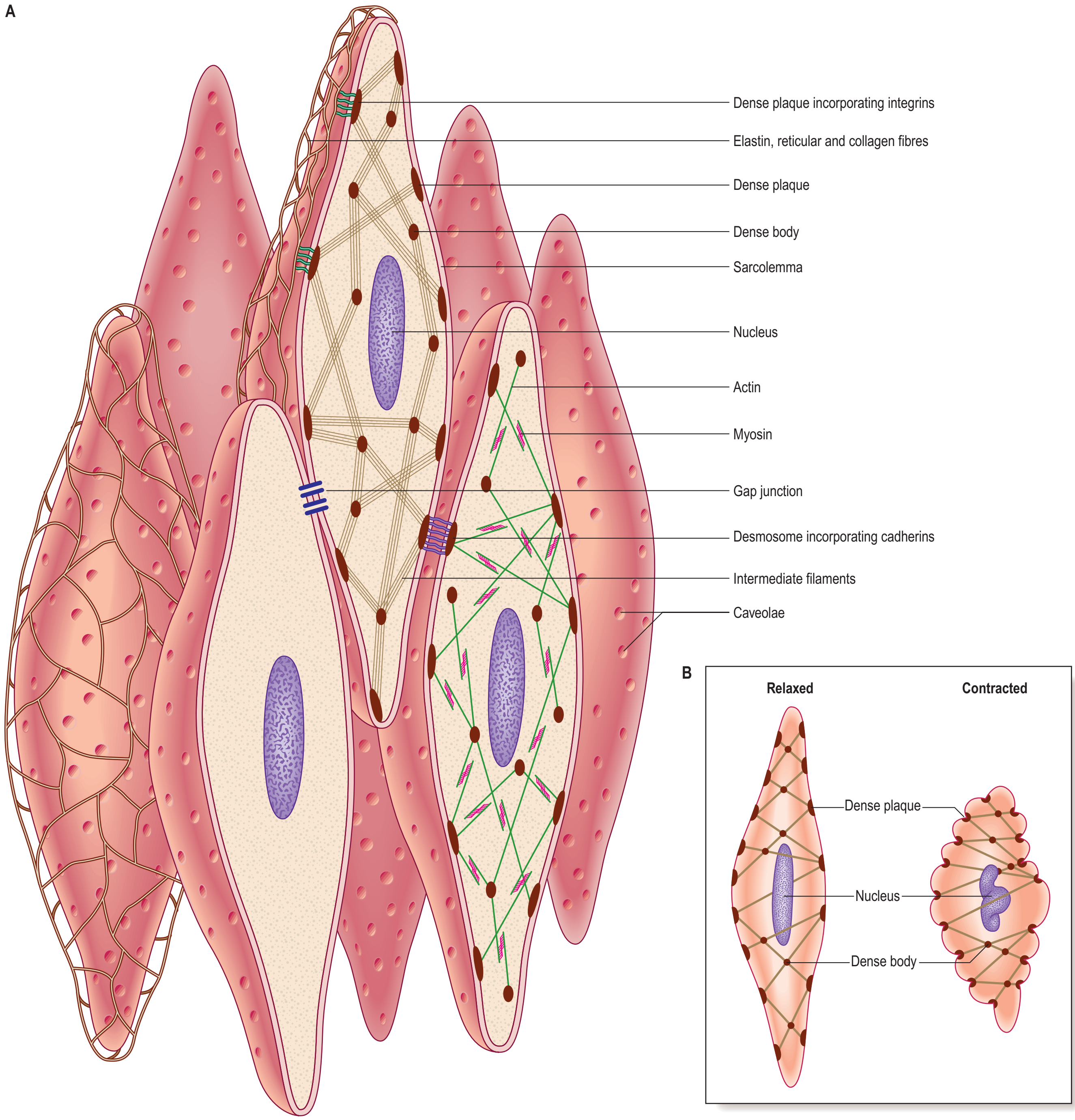

Fig. 6.3 **A**, A three-dimensional representation of smooth muscle cells. For clarity, some structural features have been separated for illustration in different cells. The spindle-shaped cells interdigitate with their long axes parallel; mechanical continuity between the cells is provided by a reticular layer of elastin and collagen fibres. The cytoskeletal framework consists of intermediate filament arrays (mainly longitudinal) and bundles of actin and myosin filaments (shown in separate cells) inserted into cytoplasmic dense bodies and submembraneous dense plaques to form a three-dimensional network. The sarcolemma contains anchoring desmosomes (adherens junctions), gap junctions and caveolae. **B**, The concertina-like change in shape of smooth muscle cells as they contract.

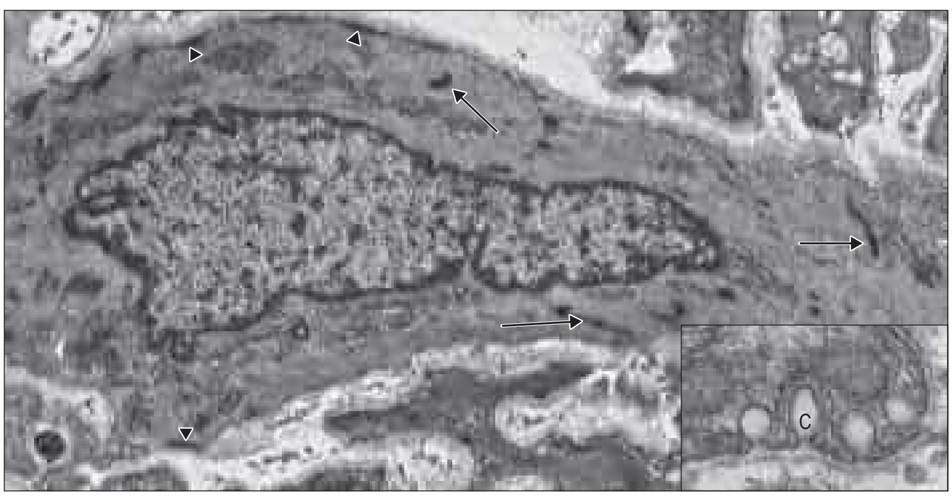

Fig. 6.4 Transmission electron micrographs showing the characteristic features of smooth muscle cells. Vascular smooth muscle in human kidney tissue, showing a cytoplasm packed densely with microfilaments (actin and myosin), cytoplasmic dense bodies (arrows) and submembraneous dense plaques (arrowheads). A basal lamina encloses the cell. The inset shows four caveolae (C) (vesicular invaginations of the cell surface at high magnification). These are associated with receptors, enzymes and ion channels important in smooth muscle function. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK. Inset courtesy of Professor Chun Y Seow, University of British Columbia.)

The membrane of smooth muscle cells contains molecules such as proteoglycans, glycoproteins and glycolipids on its exterior surface that form a thin, negatively charged network, the glycocalyx. Whilst less studied than that of endothelial cells, the smooth muscle cell glycocalyx has been shown to be important for mechanotransduction in small blood vessels. There are also a number of cell membrane-spanning proteins that extend through the glycocalyx and interact with the extracellular matrix, including cadherins and integrins. These provide structural and signalling links with the extracellular matrix and other cells, e.g. at desmosomes. The cytosolic ends of these proteins interact with components of the cytoskeleton via a complex of other proteins and kinases, and provide an important signalling pathway linking cell function to the exterior, which can modulate, for example, gene transcription and phenotype (Stegemann 2005).

Discontinuities occur in the basal lamina between adjacent cells, and here the cell membranes approach to 2–4 nm of one another to form a gap junction (see Fig. 6.2). These junctions are structurally similar to their counterparts in cardiac muscle, and are formed by binding of adjacent connexon complexes on the surface of the two cells. They provide a low-resistance channel through which electrical excitation and small molecules can pass, e.g. enabling a coordinated wave of contraction. The incidence of gap junctions varies with the anatomical site of the tissue: they appear more abundant in the type of smooth muscle that generates rhythmic (phasic) activity and they link the endothelium functionally with the underlying smooth muscle (myoendothelial gap junctions) in small resistance arteries and arterioles.

Caveolae, cup-like invaginations of the plasma membrane with a resemblance to endocytotic vesicles, are a characteristic feature of smooth muscle cells, and may form up to 30% of the membrane (see Fig. 6.4). They are associated with many receptors, ion channels, kinases and the peripheral sarcoplasmic reticulum, and may thus act as highly localized signalling microdomains. They may also act as specialized pinocytotic structures involved in fluid and electrolyte transport into the cell. Other organelles (mitochondria, ribosomes, etc.) are largely confined to the filament-free perinuclear cytoplasm, although in some smooth muscle types, including vascular smooth muscle, peripheral mitochondria, sarcoplasmic reticulum and sarcolemma form signalling microdomains.

NEUROVASCULAR SUPPLY OF SMOOTH MUSCLE

Vascular supply

The blood supply of smooth muscle is less extensive than that of striated muscle. Where the tissue is not too densely packed, afferent and efferent vessels gain access via connective tissue septa, and capillaries run in the connective tissue between small fascicles. However, unlike striated muscle, capillaries are not found in relation to individual cells.

Innervation

Smooth muscle may contract in response to nervous or hormonal stimulation, or electrical depolarization transferred from neighbouring cells. Some muscles receive a dense innervation to all cells; these are often referred to as multi-unit smooth muscles, and most blood vessels are of this type. Such innervation can precisely define contractile activity; e.g. in the iris, specific nervous control can produce either pupillary constriction or dilation. Other muscles are more sparsely innervated. They tend to display myogenic activity, initiated spontaneously or in response to stretch, which may be markedly influenced by hormones. In these muscles, which include those in the walls of the gastrointestinal tract, urinary bladder, ureter, uterus and uterine tube, innervation tends to exert a more global influence on the rate and force of intrinsically generated contractions. These muscles have been referred to as unitary smooth muscles. The terms multi-unit and unitary smooth muscles are widely used, but in practice such distinctions are better regarded as the extremes of a continuous spectrum.

Smooth muscles are innervated by unmyelinated axons whose cell bodies are located in autonomic ganglia, either in the sympathetic chain or, in the case of parasympathetic fibres, closer to the point of innervation (**Fig. 6.5**). They ramify extensively, spreading over a large area of the muscle and sending branches into the muscle fasciculi. The terminal portion of each axonal branch is beaded and consists of expanded portions, varicosities, packed with vesicles and mitochondria, separated by thin, intervaricose portions. Each varicosity is regarded as a transmitter release site and may be considered as a nerve ending in the functional sense. In this way the axonal arborization of a single autonomic neurone bears a very large number of nerve endings (up to tens of thousands), as opposed to a maximum of a few hundred in somatic motor neurones. The neuromuscular terminals of autonomic efferents are considered in more detail on page 64.

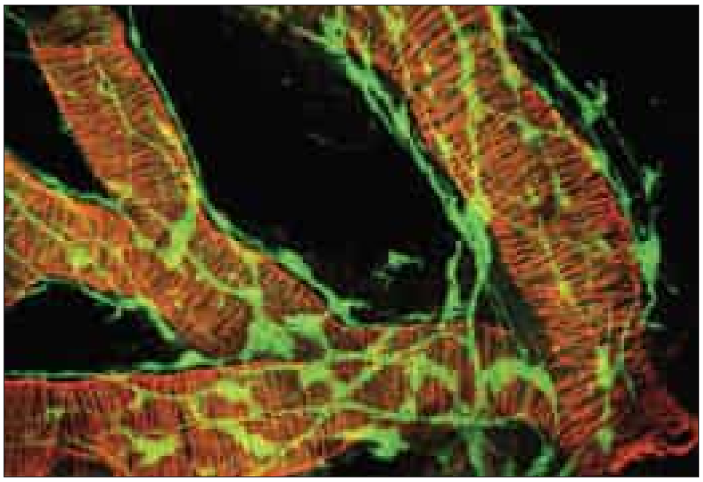

Fig. 6.5 A confocal fluorescence micrograph illustrating the innervation of airway smooth muscle in the developing human lung. Smooth muscle cells are arranged circumferentially and are labelled for actin (red); nerves and ganglia form a loose network around the smooth muscle, separated from it by up to 40 μm. Nervous tissue is labelled for PGP 9.5 (green). (Reproduced with permission from Sparrow MP, Weichselbaum M, McCray PB Jr 1999 Development of the innervation and airway smooth muscle in human fetal lung. Am J Respir Cell Mol Biol 20: 550–60.)

The neuromuscular junctions in smooth muscles do not show the consistent appearance seen in skeletal muscles. The neurotransmitter diffuses across a gap that can vary from 10 to 100 nm; even separations of up to 1 μm may still allow neuromuscular transmission to take place, although more slowly. The nerve ending is packed with vesicles but the adjacent area of the muscle cell is not structurally differentiated from that of non-junctional regions, i.e. there is no distinct synapse.

Intramuscular afferent nerves are the peripheral processes of small sensory neurones in the dorsal root ganglia. Since they are unmyelinated, contain axonal vesicles and have a beaded appearance, they are difficult to distinguish from efferent fibres, except by differential staining for neurotransmitters.

STRUCTURAL BASIS OF CONTRACTION

Although electron microscopy revealed the presence of filaments in smooth muscle some years ago, this observation alone provided little insight into their mode of function because of the lack of any obvious organization of the filaments. More recent work, using high-resolution immunocytochemistry, has revealed further details of the internal architecture of the cell and suggests a structural basis for contractile function. The model, which is illustrated in Figure 6.3, depends on the mutual interaction of two systems of filaments, one forming the cytoskeleton and the other the contractile apparatus.

Excluding the perinuclear region, the cytoplasm of a smooth muscle cell effectively consists of two structural domains. The cytoskeleton forms a structural framework that maintains the spindle-like form of the cell and provides an internal scaffold with which other elements can interact. Its major structural component is the intermediate filament desmin, with the addition of vimentin (which may also be present alone) in vascular smooth muscle. The intermediate filaments are arranged mainly in longitudinal bundles, but some filaments interconnect the bundles with each other and with the sarcolemma to form a three-dimensional network. The bundles of intermediate filaments insert into focal, electron-dense bodies, approximately 0.1 μm in diameter, which are distributed uniformly throughout the cytoplasm and also attach to dense plaques underlying the plasma membrane (see Fig. 6.3A). The cytoplasmic dense bodies and submembraneous dense plaques are equivalent to the Z-discs of striated muscle cells. They contain the actin-binding protein α-actinin and thus also anchor the actin filaments of the contractile apparatus. These form a lattice of obliquely arranged bundles throughout the cytoplasm, which transmit force to the plasma membrane and thus the basal lamina and extracellular matrix via dense plaques. These are associated with a highly structured arrangement of ancillary proteins, including vinculin and talin, In some blood vessels, notably those of the pulmonary circulation, and in the airways and probably in other smooth muscle types, there is evidence for heterogeneity of cell phenotype. The smooth muscle cells of small blood vessels and bronchioles exhibit different functional properties from those in larger vessels and airways, and may differ in morphology, expression of signalling proteins such as ion channels, and excitation–contraction coupling mechanisms. Even within individual tissues there is evidence of heterogeneity. Some myofibroblast-like cells have a function that is more secretory than contractile. The secretory phenotype is often increased in disease (e.g. chronic severe asthma, pulmonary hypertension) and is associated with increased proliferation and remodelling, and with secretion of cytokines and other mediators. Many smooth muscles seem to exhibit considerable phenotypic plasticity between these contractile and secretory phenotypes (Halayko and Solway 2001).

which in turn attach to integrins that cross the membrane and provide attachment to components of the extracellular matrix (Gunst and Zhang 2008).

An analogous arrangement underlies cell–cell attachment at desmosomes, but here the attachment between dense plaques is provided by transmembrane cadherin glycoproteins and intracellular catenins instead of integrins and talin. Mechanical deformation of the cell may be linked to cell signalling mechanisms via focal adhesion kinase (FAK) and its substrate paxillin; phosphorylation of talin and paxillin may modulate the deformability of the smooth muscle cell. Other regulatory proteins also associate specifically with actin, such as caldesmon and calponin. The cytoskeleton is not a passive structure. It adapts dynamically to load and is modulated by cell surface receptors including integrins and agonist binding to G-protein coupled receptors, and so contributes to contraction (Gunst and Zhang 2008). This presumably contributes to the low energy requirements of smooth muscle contraction because dynamic reorganization of the cytoskeleton following active contraction allows cell shortening to be maintained without further energy expenditure.

The ratio of actin to myosin is about eight times greater in smooth compared to striated muscle, reflecting the greater length of actin filaments in smooth muscle. Smooth muscle myosin filaments are 1.5–2 μm long, somewhat longer than those of striated muscle. Although smooth muscle cells contain less myosin, the longer filaments are capable of generating considerable force. The myosin filaments of smooth muscle are also assembled differently, such that their head regions lie symmetrically on either side of a ribbon-like filament, rather than imposing a bipolar organization on the filament. Actin filaments, to which they bind, can thus slide along the whole length of the myosin filament during contraction. In addition, dynamic polymerization of both myosin and actin monomers during activation can alter the length of the contractile filaments. These differences underpin the ability of smooth muscle to undergo much greater changes in length than striated muscle. Actin–myosin filament sliding generates tension, which transmits to focal regions of the plasma membrane, changing the cell to a shorter, more rounded shape (see Fig. 6.3B) and often deforming the nucleus to a corkscrew-like profile.

Although some smooth muscles can generate as much force per unit cross-sectional area as skeletal muscle, the force always develops much more slowly than in striated muscle. Smooth muscle can contract by more than 80%, a much greater range of shortening than the 30% or so to which striated muscle is limited. The significance of this property is illustrated by the urinary bladder, which is capable of emptying completely from an internal volume of 300 ml or more. Smooth muscles can maintain tension for long periods with very little expenditure of energy. Many smooth muscle structures are able to generate spontaneous contractions; examples are found in the walls of the intestines, ureter and uterine tube.

EXCITATION–CONTRACTION COUPLING IN SMOOTH MUSCLE

Excitation–contraction coupling in smooth muscle is more complex than in skeletal or cardiac muscle, and may be electromechanical or pharmacomechanical. Electromechanical coupling involves depolarization of the cell membrane by an action potential, and may be generated when a membrane receptor, usually linked with an ion channel, is occupied by a neurotransmitter, hormone or other blood-borne substance. It is most commonly seen in unitary and phasic smooth muscles such as those of the viscera, with transmission of electrical excitation from cell to cell via gap junctions. In some types of smooth muscle depolarization may be the consequence of other stimuli, such as cooling, stretch and even light.

Pharmacomechanical coupling is a receptor-mediated and G-protein coupled process, and is the major mechanism in tonic smooth muscles such as in the vasculature and airways. It may involve several pathways, including formation of inositol trisphosphate, which triggers intracellular calcium release from the sarcoplasmic reticulum, activation of voltage-independent calcium channels in the sarcolemma, and depolarization causing activation of voltage-dependent calcium channels (reviewed in Berridge (2008)). In addition, many receptors also couple to kinases that modulate contraction in a calcium-independent fashion, either via myosin phosphatase (see below) or via the actin cytoskeleton (see above). The extent to which any of these pathways contributes to activation varies between different types of smooth muscle.

Whilst the regulation of contraction of smooth muscle is largely calcium-dependent, in contrast to cardiac and skeletal muscle, the effects of an elevation of intracellular calcium are mediated via myosin rather than actin/tropomyosin. Most smooth muscles contain little or no troponin, and instead calcium binds to calmodulin. The calcium–calmodulin complex regulates the activity of myosin light chain kinase, which phosphorylates myosin regulatory light chains and initiates the myosin–actin adenosine 5′-triphosphatase (ATPase) cycle. The enzymatic activation process is therefore inherently slow. Myosin phosphatase dephosphorylates myosin and thus promotes relaxation. The degree of myosin phosphorylation and therefore contraction depends on the relative activities of myosin light chain kinase and myosin phosphatase. Thus inhibition of the phosphatase, e.g. by Rho kinase, increases phosphorylation for any level of calcium (i.e. increases calcium sensitivity). The latter is a central component of the response to many constrictor agonists.

Regulation of smooth muscle intracellular calcium

Intracellular calcium is a key determinant of smooth muscle function, including contraction and also proliferation, migration and secretion of mediators. Its regulation in smooth muscle is particularly complex, and involves calcium entry via both voltage-dependent and independent ion channels, release from and reuptake into intracellular stores such as the sarcoplasmic reticulum, and modulation by mitochondria.

ORIGIN OF SMOOTH MUSCLE

It was thought that all smooth muscle cells developed *in situ* exclusively from the splanchnopleuric mesenchyme in the walls of the anlagen of the viscera and around the endothelium of blood vessels. However, endothelial and tunica media smooth muscle cells arise from the epithelial plate of individual somites (Scaal and Christ 2004), and the smooth muscle of the sphincter and dilator pupillae is derived from neurectoderm.

Following a period of proliferation, clusters of myoblasts become elongated in the same orientation. Dense bodies, associated with actin and cytoskeletal filaments, appear in the cytoplasm, and the surface membrane starts to acquire its specialized features, i.e. caveolae, adherens junctions and gap junctions. Cytoskeletal filaments extend to insert into the submembraneous dense plaques and cytoplasmic dense bodies. Thick filaments are seen a few days after the first appearance of thin filaments and intermediate filaments, and from this time the cells are able to contract. During development, dense bodies increase in number and further elements of the cytoskeleton are added. In addition to synthesizing the cytoskeleton and contractile apparatus, the differentiating cells express and secrete components of the extracellular matrix.

In a developing smooth muscle, all the cells express characteristics of the same stage of differentiation and there are no successive waves of differentiation. From its earliest appearance to maturity, a smooth muscle increases several hundred-fold in mass, partly by a 2–4-fold increase in the size of individual cells, but mainly by a very large increase in cell number. Growth occurs by division of cells in every part of the muscle, not just at its surface or ends. Mitosis occurs in cells in which differentiation is already well advanced, as evidenced by the presence of myofilaments and membrane specializations. Mitotic smooth muscle cells may be found at any stage of life but their numbers peak before birth, at a time that differs for different muscles; they are rare in the adult unless the tissue is stimulated to hypertrophy (as in the pregnant uterus) or to repair. The ability of mature cells to undergo mitosis therefore differs between the three major types of muscle: skeletal muscle cells cannot divide at all after differentiation; cardiac muscle cells can divide but only before birth; and smooth muscle cells appear to remain capable of division throughout life.

During the early stages of development, smooth muscle expresses embryonic and non-muscle isoforms of myosin. The proportions of these isoforms decrease progressively. Initially, SM-1 is the dominant or exclusive smooth muscle heavy chain isoform and the SM-2 isoform becomes more established later. For a review of the development of vascular smooth muscle, see Owens et al (2004).

SMOOTH MUSCLE REMODELLING IN DISEASE

The ability of smooth muscle cells to divide and change phenotype throughout life means that smooth muscle has significant plasticity and can adapt to changing needs and stimuli, e.g. in the walls of the uterus during pregnancy. This can, however, have detrimental consequences in disease.

The arrangement of peripheral signalling microdomains (see above) is vital to this process. Sequential release from inositol trisphosphate receptors and calcium-sensitive ryanodine receptors (calcium release channels) in the sarcoplasmic reticulum, together with reuptake via the sarco-endoplasmic reticulum calcium ATPase (SERCA), give rise to slow calcium oscillations; it is now thought that the rate of oscillations, rather than the average level of calcium, is the prime driver for smooth muscle function, including contraction and activation of gene transcription (Berridge 2008).

Sustained stress (physical or oxidative), tissue damage, inflammatory mediators and other stimuli can promote enhanced growth, apoptosis (programmed cell death, p. 26) and switching to a more secretory phenotype (see above). Increased mediator release as a result of switching to a more secretory phenotype can potentiate such changes and attract inflammatory cells, resulting in a positive feedback loop. Chronic disease and inflammation can therefore lead to extensive smooth muscle remodelling, a major contributing factor to, for example, chronic asthma and pulmonary hypertension, which significantly worsens the conditions (Mahn et al 2010, Stenmark et al 2006). Peripheral vascular remodelling may also occur in essential hypertension and diabetes.

CARDIOVASCULAR AND LYMPHATIC SYSTEMS

GENERAL ORGANIZATION

Cells of peripheral blood, suspended in plasma, circulate through the body in the blood vascular system. Fluid and solutes exchange between the plasma and interstitium across capillaries and small venules. Excess interstitial fluid from peripheral tissues returns to the blood vascular system via the lymphatic system, which also provides a channel for the migration of leukocytes and the absorption of certain nutrients from the gut.

The cardiovascular system carries nutrients, oxygen, hormones, etc. throughout the body and the blood redistributes and disperses heat. As a consequence of the hydrostatic pressure, the system also has mechanical effects, such as maintaining tissue turgidity. Blood circulates within a system made up of the heart, the central pump and main motor of the system; arteries, which lead away from the heart and carry the blood to the periphery; and veins, which return the blood to the heart. The heart is essentially a pair of muscular pumps, one feeding the pulmonary circulation, which is responsible for gas exchange in the lungs and has a low hydrostatic pressure, and the other feeding the systemic circulation, which has a high hydrostatic pressure and serves the rest of the body. With limited exceptions, both circulations form a closed system of tubes, so that blood *per se* does not usually leave the circulation.

From the centre to the periphery, the vascular tree shows three main modifications. The arteries increase in number by repeated bifurcation and by sending out side branches, in both the systemic and the pulmonary circulation. For example, the aorta, which carries blood from the heart to the systemic circulation, gives rise to about 4×10^6 arterioles and four times as many capillaries. The arteries also decrease in diameter, although not to the same extent as their increase in number, so that a hypothetical cross-section of all the vessels will show an increase in total area with increasing distance from the heart. At its emergence from the heart, the aorta of an adult man has an outer diameter of approximately 30 mm (cross-sectional area of nearly 7 cm^2). The diameter decreases along the arterial tree until it is as little as 10 μm in arterioles (each with a cross-sectional area of about 80 μm^2). However, given the enormous number of arterioles, the total cross-sectional area at this level is approximately 150 cm^2, more than 200 times that of the aorta. As a result, blood flow is faster near the heart than at the periphery.

The walls of arteries decrease in thickness towards the periphery, although this is not as substantial as the reduction in vessel diameter. Consequently, in the smallest arteries (arterioles), the thickness of the wall represents about half the outer radius of the vessel, whereas in a large vessel it represents between one-fifteenth and one-fifth, e.g. in the thoracic aorta the radius is approximately 17 mm and the wall thickness 1.1 mm.

Venules, which return blood from the capillaries, converge on each other, forming a progressively smaller number of veins of increasingly large size. As with arteries, the hypothetical total cross-sectional area of all veins at a given level reduces nearer to the heart. Eventually, only the two largest veins, the superior and inferior venae cavae, open into the heart from the systemic circulation. A similar pattern is found in the pulmonary circulation but here the vascular loop is shorter and has fewer branch points, and consequently, the number of vessels is smaller than in the systemic circulation. The total end-to-end length of the vascular network in a typical adult is twice the circumference of the earth.

Large arteries, such as the thoracic aorta and subclavian, axillary, femoral and popliteal arteries, lie close to a single vein that drains the same territory as that supplied by the artery. Other arteries are usually flanked by two veins, satellite veins (venae comitantes), which lie on either side of the artery and have numerous cross-connections; the whole is enclosed in a single connective tissue sheath. The artery and the two satellite veins are often associated with a nerve, and when they are surrounded by a common connective tissue sheath they form a neurovascular bundle.

The close association between the larger arteries and veins in the limbs allows counterflow exchange of heat. This mechanism promotes heat transfer from arterial to venous blood and thus helps to preserve body heat. Counterflow heat exchange systems are found in certain organs, e.g. in the testis, where the pampiniform plexus of veins surrounds the testicular artery (this arrangement not only conserves body heat, but also maintains the temperature of the testis below average body temperature). Counterflow exchange mechanisms are found in the microcirculation, as in the arterial and venous sinusoids of the vasa recta in the renal medulla. Here, countercurrent exchange retains solutes at a high concentration in the medullary interstitium, with efferent venous blood transferring solutes to the afferent arterial supply; this mechanism is essential for concentration of the urine.

Arteries and veins are named primarily according to their anatomical position. In functional terms, four main classes of vessel are described: conducting and distributing vessels (large arteries), resistance vessels (small arteries but mainly arterioles), exchange vessels (capillaries, sinusoids and small venules) and capacitance vessels (veins). Structurally, arteries can also be divided into elastic and muscular types. Although muscle cells and elastic tissue are present in all arteries, the relative amount of elastic material is greatest in the largest vessels, whereas the relative amount of smooth muscle increases progressively towards the smallest arteries.

The large conducting arteries that arise from the heart, together with their main branches, are characterized by the predominantly elastic properties of their walls. Distributing vessels are smaller arteries supplying the individual organs, and their walls are characterized by a well-developed muscular component. Resistance vessels include the smallest arteries and arterioles, and are highly muscularized. They provide the major part of peripheral resistance to blood flow and so cause the largest drop in blood pressure before the blood flows into the tissue capillary beds.

Capillaries, sinusoids and small (postcapillary) venules are collectively termed exchange vessels. Their thin walls allow exchange between blood and the interstitial fluid that surrounds all cells: this is the essential function of a circulatory system. Arterioles, capillaries and venules constitute the microvasculature, the structural basis of the microcirculation.

Larger venules and veins form an extensive but variable, large-volume, low-pressure system of vessels conveying blood back to the heart. Their high capacitance is due to the significant distensibility (compliance) of their walls, so that the content of blood is high even at low pressures. Veins contain the greatest proportion of blood, reflecting their large relative volume.

Blood from the gastrointestinal tract (with the exception of the lower part of the anal canal) and from the spleen, pancreas and gallbladder drains to the liver via the portal vein. The portal vein ramifies within the substance of the liver like an artery and ends in the hepatic sinusoids. These drain into the hepatic veins, which in turn drain into the inferior vena cava. Blood supplying the abdominal organs thus passes through two sets of capillaries before it returns to the heart. The first provides the organs with oxygenated blood, and the second carries deoxygenated blood, rich in absorption products from the intestine, through the liver parenchyma. A venous portal circulation also connects the median eminence and infundibulum of the hypothalamus with the adenohypophysis. In essence, a venous portal system is a capillary network that lies between two veins, instead of between an artery and a vein, which is the more usual arrangement in the circulation. A capillary network may also be interposed between two arteries, e.g. in the renal glomeruli, where the glomerular capillary bed lies between afferent and efferent arterioles. This maintains a relatively high-pressure system, which is important for renal filtration.

A parallel circulatory system is provided by the lymphatic vessels and lymph nodes. Lymphatic vessels originate in peripheral tissues as blind-ended endothelial tubes that collect excess fluid from the interstitial spaces between cells and conduct it as lymph. Lymph is returned to the blood vascular system via lymphatic vessels, which converge on the large veins in the root of the neck.

The development of blood vessels is described in Chapter 13.

General features of vessel walls

All blood vessels, with the exception of capillaries and venules, have walls consisting of three concentric layers (tunicae) (see Fig. 6.8). The intima (tunica intima) is the innermost layer. Its main component, the endothelium, lines the entire vascular tree, including the heart and lymphatic vessels. The media (tunica media) contains muscle cells, elastic fibres and collagen. While it is by far the thickest layer in arteries, the media is absent in capillaries and is comparatively thin in veins. The adventitia (tunica adventitia) is the outer coat of the vessel and consists of connective tissue, nerves and vessel capillaries (vasa vasorum). It links the vessels to the surrounding tissues. Vessels differ in the relative thicknesses and detailed compositions of these layers.

Large elastic arteries

The aorta and its largest branches (brachiocephalic, common carotid, subclavian and common iliac arteries) are large elastic arteries that conduct blood to the medium-sized distributing arteries.

The intima is made of an endothelium, resting on a basal lamina, and a subendothelial connective tissue layer. The endothelial cells are flat, elongated and polygonal in outline, with their long axes parallel to the direction of blood flow (see Fig. 6.17). The subendothelial layer is well developed, contains elastic fibres and type I collagen fibrils, fibroblasts and small, smooth muscle-like myointimal cells. The latter accumulate lipid with age, and in an extreme form this contributes to atherosclerosis. Thickening of the intima progresses with age and is more marked in the distal than in the proximal segment of the aorta.

A prominent internal elastic lamina, sometimes split, lies between intima and media. This lamina is smooth, measures about 1 μm in thickness, and, with the elastic lamellae of the media, is stretched under the effect of systolic pressure, recoiling elastically in diastole. Elastic arteries can thus sustain continuous blood flow despite the pulsatile cardiac output, and smooth out the cyclical pressure wave. The media has a markedly layered structure, in which fenestrated layers of elastin (elastic lamellae) alternate with interlamellar smooth muscle cells (**Fig. 6.6**), collagen and fine elastic fibres. The arrangement is very regular, such that each elastic lamella and adjacent interlamellar zone is regarded as a 'lamellar unit' of the media. In the human aorta there are approximately 52 lamellar units, measuring about 11 μm in thickness. Number and thickness of lamellar units increases during postnatal development, from 40 at birth.

The adventitia is well developed. In addition to collagen and elastic fibres, it contains flattened fibroblasts with extremely long, thin processes, macrophages and mast cells, nerve bundles and lymphatic vessels. The vasa vasorum is usually confined to the adventitia.

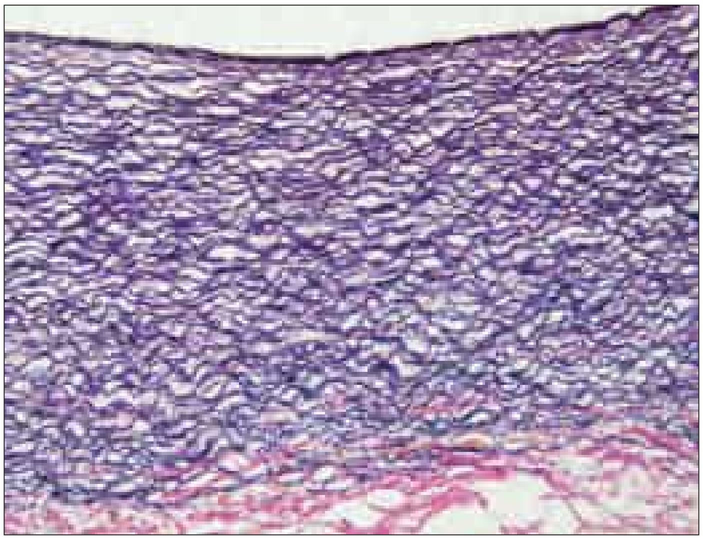

Fig. 6.6 Elastic artery (human aorta), stained for elastic fibres. The dense staining of the internal elastic lamina is seen close to the luminal surface (top); elastic lamellae fill the tunica media and merge with the external elastic lamina at its junction with the collagenous adventitia (red fibres, below). Compare with Figure 6.20. van Gieson stain. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

Muscular arteries

Muscular arteries are characterized by the predominance of smooth muscle in the media (**Fig. 6.7**). The intima consists of an endothelium, similar to that of elastic arteries, which rests on a basal lamina and subendothelial connective tissue. The internal elastic lamina (see Fig. 6.7; **Fig. 6.8**) is a distinct, thin layer, sometimes duplicated and occasionally absent. It is thrown into wavy folds as a result of contraction of smooth muscle in the media. Some 75% of the mass of the media consists of smooth muscle cells that run spirally or circumferentially around the vessel wall. The relative amount of extracellular matrix is therefore less than in large arteries, although fine elastic fibres that run mainly parallel to the muscle cells are present. An external elastic lamina, composed of sheets of elastic fibres, forms a less compact layer than the internal lamina, and separates the media from the adventitia in larger muscular arteries. The adventitia is made of fibroelastic connective tissue, and can be as thick as the media in the smaller arteries. The inner part of the adventitia contains more elastic than collagen fibres.

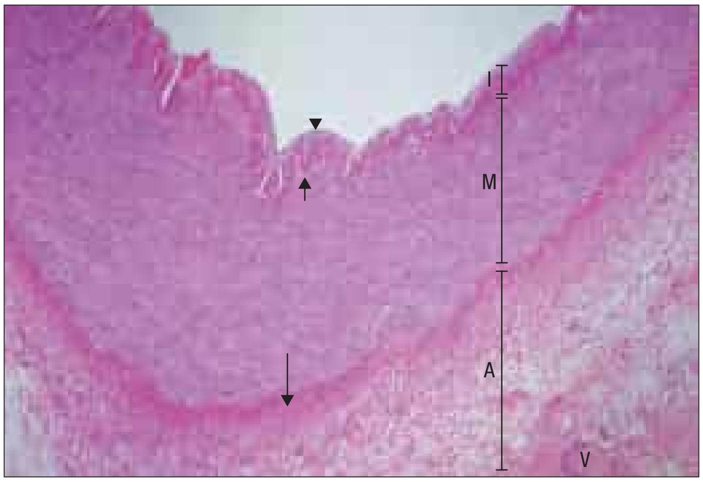

Fig. 6.7 The wall of a human muscular artery. The intima (I) forms the innermost layer, lined by an endothelium (arrowhead) and separated from the middle muscular layer, the media (M), by an internal elastic lamina (short arrow). A more diffuse external elastic lamina (long arrow) divides the media from the outermost collagenous adventitia (A), within which lie the vasa vasorum (V). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

Arterioles

In arterioles (**Figs 6.9–6.10**), the endothelial cells are smaller than in large arteries, but their nuclear region is thicker and often projects markedly into the lumen. The nuclei are elongated and orientated parallel to the vessel length, as is the long axis of the cell. The basal surface of the endothelium contacts a basal lamina, but an internal elastic lamina is either absent or highly fenestrated and traversed by cytoplasmic processes of muscle or endothelial cells. The muscle cells are larger in cytoplasmic volume than those in the walls of large arteries and form a layer one or two cells thick. They are arranged circumferentially and are tightly wound around the endothelium. In the smallest arterioles each cell makes several turns, producing extensive apposition between parts of the same cell. Contraction of the muscle constricts the lumen, and so controls blood flow into the capillary bed; arterioles thus act functionally as precapillary sphincters, even though the absence of an anatomically delineated sphincter means that the term is no longer commonly used. Contraction of arterioles is primarily regulated by local vasoactive and metabolic factors, but can also be controlled by central mechanisms.

Arterioles are usually densely innervated by sympathetic fibres, via small bundles of varicose axons packed with transmitter vesicles, mostly of the adrenergic type. The distance between axolemma and muscle cell membrane can be as little as 50–100 nm and the gap is occupied only by a basal lamina. Autonomic neuromuscular junctions are very common in arterioles. Arteriolar adventitia is very thin.

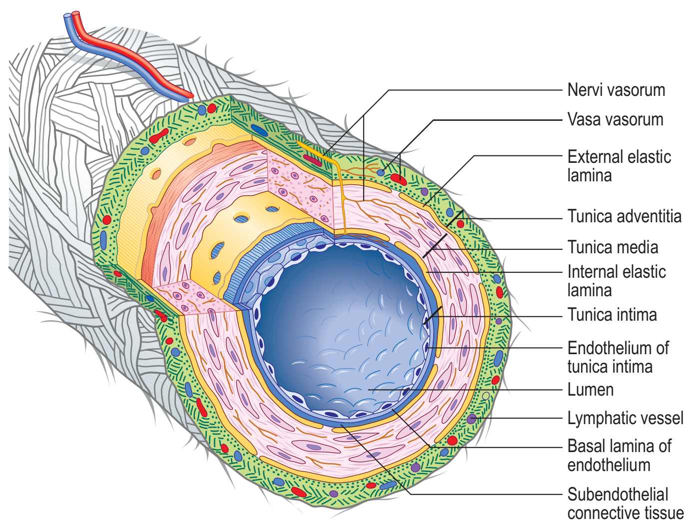

Fig. 6.8 The principal structural features of the larger blood vessels as seen in a muscular artery.

Capillaries

The capillary wall (**Fig. 6.11**) is formed by an endothelium and its basal lamina, plus a few isolated pericytes. Capillaries are the vessels closest

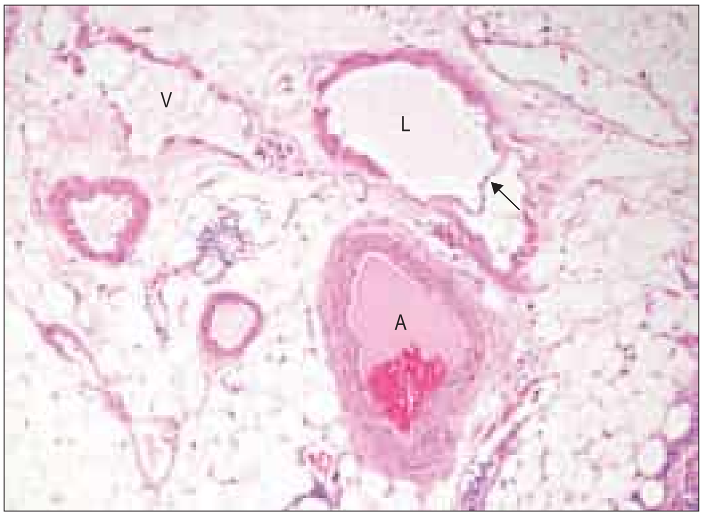

Fig. 6.9 An arteriole (A) and accompanying venule (V) and lymphatic vessel (L) (with a valve, arrow) in adipose tissue around a lymph node (human). Note the relative thicknesses of the vessel walls, in comparison with the diameters of their lumens. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

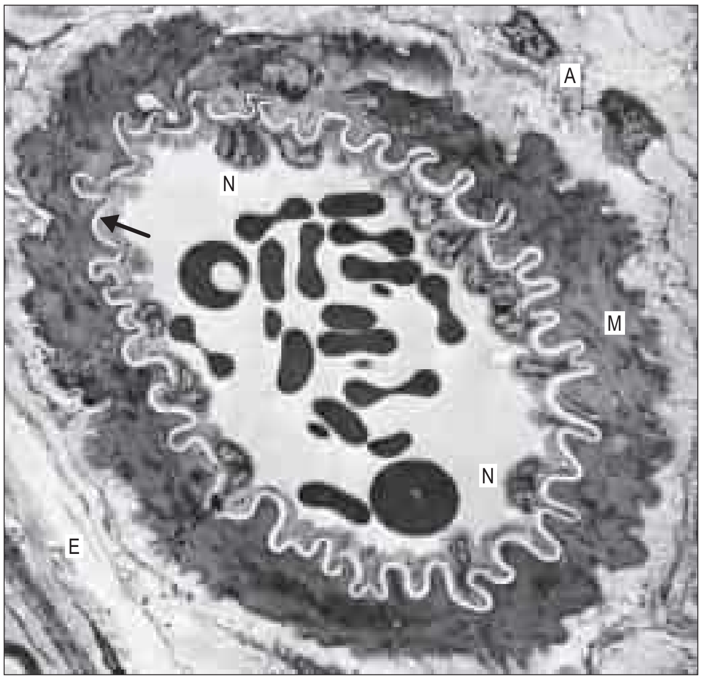

Fig. 6.10 A transmission electron micrograph of a small arteriole in the epineurium of a peripheral nerve. The vessel lumen contains erythrocytes and is lined by endothelial cells (with nuclei, N, projecting into the lumen); note the electron-lucent internal elastic lamina (pale, wavy line, arrow), the media containing densely filamentous smooth muscle cells (M) and the connective tissue of the adventitia (A) merging with that of the epineurium (E). (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

to the tissue they supply and their wall constitutes a minimal barrier between blood and the surrounding tissues. Capillary structure varies in different locations. Capillaries measure 4–8 μm in diameter (much more in the case of sinusoids) and are hundreds of microns long. Their lumen is just large enough to admit the passage of single blood cells, usually with considerable deformation. Typically a single endothelial cell forms the wall of a capillary, and there are junctional complexes between extensions of the same cell.

Endothelial cells are joined by tight junctions (occluding junctions, zonulae adherentes), forming a diffusion barrier. Capillary permeability varies greatly among tissues and is correlated largely with the type of endothelium. The majority of tissues, including brain, muscle, lung and connective tissues, contain capillaries with a continuous, unbroken layer of endothelium (continuous endothelium). This is impermeable to proteins, although electrolytes can diffuse through the tight junc-

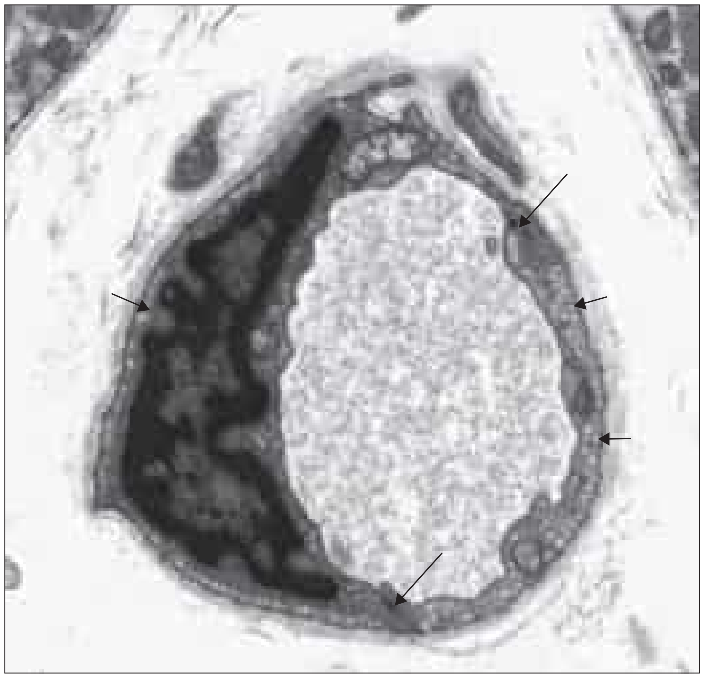

Fig. 6.11 Transmission electron micrograph of a capillary in a human muscle biopsy specimen. An endothelial cell with its nucleus in the plane of section forms adherens junctions (long arrows) with either a second cell or an extension of itself. The cytoplasm contains numerous transcytotic vesicles (short arrows). A basal lamina surrounds the capillary. (Courtesy of Dr Bart Wagner, Histopathology Department, Sheffield Teaching Hospitals, UK.)

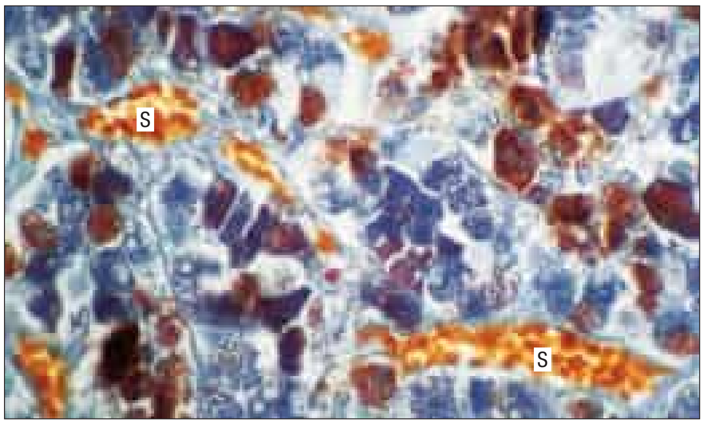

Fig. 6.12 Expanded sinusoids (S), typical of endocrine glands and certain other tissues, are seen here containing erythrocytes (orange) in the adenohypophysis. Endocrine cells stain either blue or reddish-brown, or are poorly stained in this trichrome preparation.

tions, albeit relatively slowly. Their passage is further limited in brain, thymic cortex and testis by particularly tight junctions.

Endothelial cells of some capillaries have fenestrations, or pores, through their cytoplasm, which facilitate diffusion. Fenestrations are approximately circular and 50–100 nm in diameter, and at their edge the luminal and abluminal membranes of the endothelial cell come into contact. The fenestration itself is usually occupied by a thin, electron-dense diaphragm containing glycoprotein PV-1, which restricts passage of large molecules such as proteins. Fenestrated capillaries occur in intestinal mucosae, endocrine and exocrine glands, and renal glomeruli, where they may lack a diaphragm. Fenestrations are almost invariably present in capillaries that lie close to an epithelium, including the skin.

Sinusoids

Sinusoids are expanded capillaries (Fig. 6.12), and are large and irregular in shape. They have true discontinuities in their walls, allowing intimate contact between blood and the parenchyma. The discontinuities are formed by gaps between fenestrated endothelial cells, such that the sinusoidal lining, and sometimes also the basal lamina, is incomplete. Sinusoids occur in large numbers in the liver (where a basal lamina is completely absent), spleen, bone marrow, adenohypophysis (see Fig. 6.12) and suprarenal medulla.

Venules

When two or more capillaries converge, the resulting vessel is larger (10–30 μm) and is known as a venule (postcapillary venule). Venules (see Fig. 6.9) are essentially tubes of flat, oval or polygonal endothelial cells surrounded by basal lamina and, in the larger vessels, by a delicate adventitia of a few fibroblasts and collagen fibres mainly running longitudinally. Pericytes (see Fig. 6.21) support the walls of these venules.

Postcapillary venules are sites of leukocyte migration. In venules of mucosa-associated lymphoid tissue (MALT), particularly of the gut and bronchi, and in the lymph nodes and thymus, endothelial cells are taller and have intercellular junctions through which lymphocytes and other blood components can readily pass. These are known as high endothelial venules (HEVs) (see Figs 6.18–6.19). Elsewhere, venules are believed to be a major site where migration of neutrophils, macrophages and other leukocytes into extravascular spaces occurs, and where neutrophils may temporarily attach, forming marginated pools.

In general, the endothelium of venules has few tight junctions and is relatively permeable. The intercellular junctions of venules are sensitive to inflammatory agents, which increase their permeability to fluids and defensive cells, and facilitate leukocyte extravasation by diapedesis.

Venules do not acquire musculature until they are about 50 μm in outer diameter, when they are known as muscular venules. This distinction is important because postcapillary venules, which lack muscle in their walls, are as permeable to solutes as capillaries and are thus part of the microcirculatory bed. At the level of the postcapillary venule the cross-sectional area of the vascular tree is at its maximum, and there is a dramatic fall in pressure (from 25 mmHg in the capillary to approximately 5 mmHg). Muscular venules converge to produce a series of veins of progressively larger diameter. Venules and veins are capacitance vessels, i.e. they have thin distensible walls that can hold a large volume of blood and accommodate luminal pressure changes.

Fluid exchange in the microvasculature

The microvasculature is important for the creation and maintenance of the interstitial fluid that bathes the cells. The thin walls of capillaries and small venules allow easy diffusion of fluid and most small molecules, but the endothelial barrier prevents movement of proteins; consequently, plasma and interstitial fluid have almost identical compositions, except that the latter contains very little protein. Fluid transfer across these exchange vessels is driven by the balance between the hydrostatic pressure (i.e. blood pressure within them) forcing fluid into the tissues, and the oncotic pressure (colloidal osmotic pressure, reflecting the difference in protein concentrations) drawing fluid back into the vessels. These 'Starling forces' are normally closely balanced, so differences in hydrostatic pressure mean that fluid tends to be filtered into the tissue at the arterial side of the exchange vessels and largely reabsorbed at the venous side. The balance forms lymph. Disruption of the balance (e.g. high venous pressure in the feet) can lead to accumulation of tissue fluid (oedema) and swelling.

Inflammation and endothelial permeability

Inflammatory mediators increase the permeability of capillaries and small venules by causing contraction of endothelial cells and so loosening tight junctions. This facilitates leukocyte extravasation by diapedesis but also disrupts normal barrier function, allowing extravasation of protein and fluids. The consequence is tissue oedema and the swelling that is commonly associated with inflammation. There is considerable cross-talk between endothelial cells and cells of the immune system.

Veins

Veins are characterized by a relatively thin wall in comparison to arteries of similar size and by a large capacitance. Wall thickness is not correlated exactly with the size of the vein and varies in different regions, e.g. the wall is thicker in veins of the leg than it is in veins of a similar size in the arm.

The structural plan of the wall is similar to that of other vessels, except that the amount of muscle is considerably less than in arteries, while collagen and, in some veins, elastic fibres predominate. In most veins, e.g. those of the limbs, the muscle is arranged approximately circularly. Longitudinal muscle is present in the iliac, brachiocephalic, portal and renal veins and in the superior and inferior venae cavae. Muscle is absent in the maternal placental veins, dural venous sinuses, pial and retinal veins, veins of trabecular bone and the venous spaces of erectile tissue; these veins consist of endothelium supported by variable amounts of connective tissue. Distinction between the media and adventitial layers is often difficult, and a discrete internal elastic lamina is absent.

Tethering of some veins to connective tissue fasciae and other surrounding tissues may prevent collapse of the vessel even under negative pressure (e.g. in the cranium). Pressure within the venous system does not normally exceed 5 mmHg, and it decreases as the veins grow larger and fewer in number, approaching zero close to the heart. As they contain only a small amount of muscle and commonly have a large calibre, veins have limited influence on blood flow. However, venoconstriction is an important component of the baroreceptor reflex because it reduces vein compliance and therefore capacity, effectively mobilizing blood to maintain or increase central venous pressure, and hence cardiac output. A sudden fall in blood volume, e.g. following a haemorrhage, also initiates the elastic recoil and reflex constriction of veins, in order to compensate for the blood loss and maintain central venous pressure and venous return to the heart. Vasoconstriction in cutaneous veins in response to cooling is important in thermoregulation.

Most veins have valves to prevent reflux of blood (**Figs 6.13–6.14**). A valve is formed by an inward projection of the intima, strengthened by collagen and elastic fibres, and covered by endothelium that differs in orientation on its two surfaces. Surfaces facing the vessel wall have transversely arranged endothelial cells, whereas on the luminal surface of the valve, over which the main stream of blood flows, cells are arranged longitudinally in the direction of flow. Most commonly two, or occasionally three, valves lie opposite one another; sometimes only one is present. They are found in small veins or where tributaries join

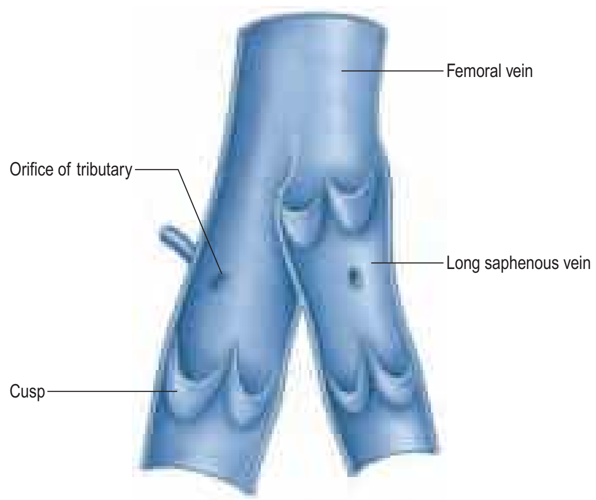

Fig. 6.13 The upper portions of the femoral and long saphenous veins laid open to show the valves, at about two-thirds of their natural size.

Fig. 6.14 A valve in a human small vein, formed from flap-like extensions of the intima that close when pressure increases on the proximal side, preventing backflow. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

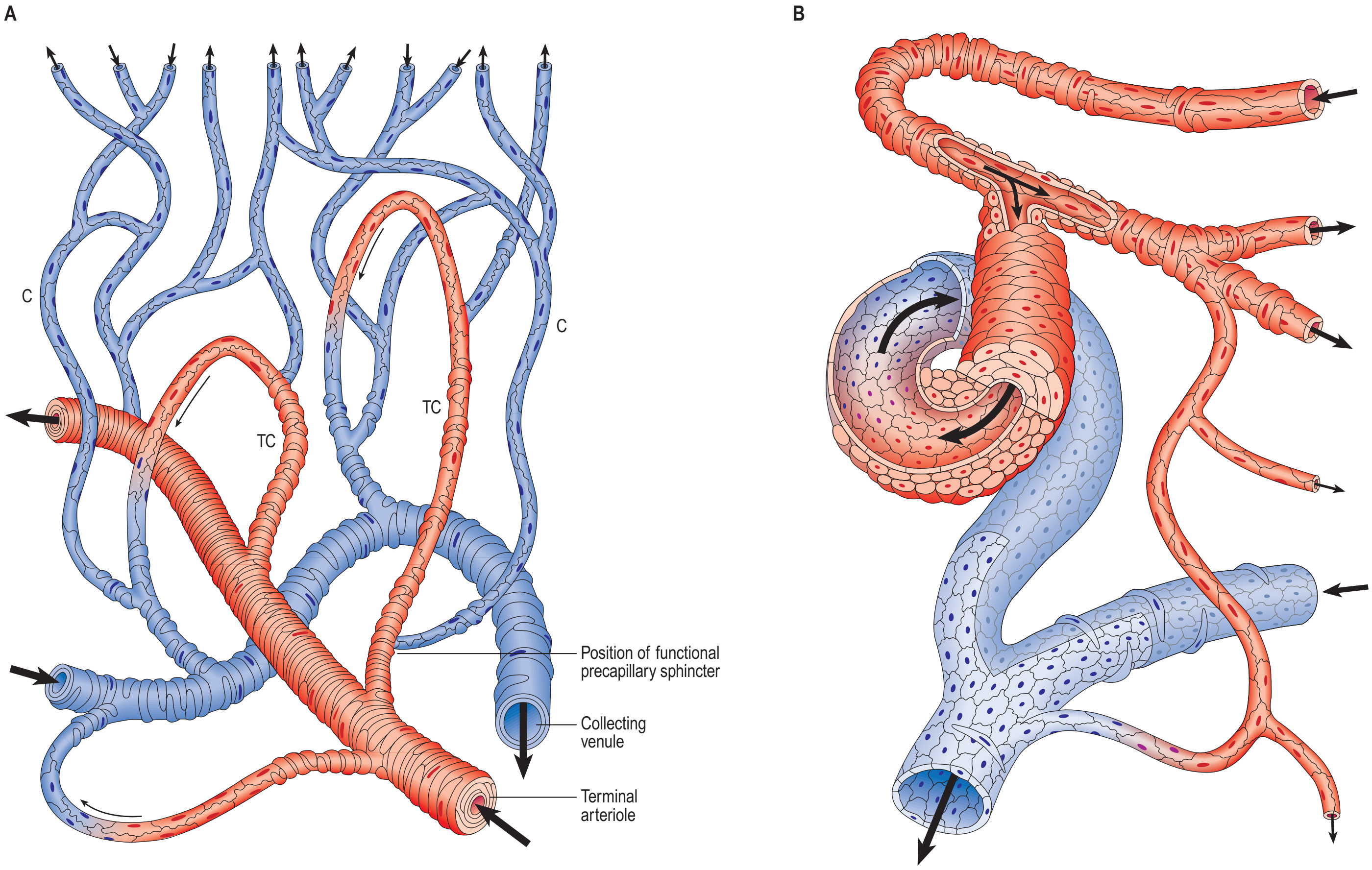

Fig. 6.15 **A**, A microcirculatory unit, showing a terminal arteriole, thoroughfare channels (TC) formed of metarterioles, capillaries (C) and collecting venule. The distribution of smooth muscle cells and one of the precapillary sites where perfusion of the capillary bed is regulated are also shown. **B**, An arteriovenous anastomosis. Note the thick wall of the anastomotic channel composed of layers of modified smooth muscle cells.

larger veins. The valves are semilunar (cusps) and attached by their convex edges to the venous wall. Their concave margins are directed with the flow and lie against the wall as long as flow is towards the heart. When blood flow reverses, the valves close and blood fills an expanded region of the wall, a sinus, on the cardiac side of the closed valve. This may give a 'knotted' (varicose) appearance to the distended veins, if these have many valves. In the limbs, especially the legs where venous return is against gravity, valves are of great importance in aiding venous flow. Blood is moved towards the heart by the intermittent pressure produced by contractions of the surrounding muscles (the muscle pump). Valves are absent in the veins of the thorax and abdomen.

VASCULAR SHUNTS AND ANASTOMOSES

Arteriovenous shunts and anastomoses

Communications between the arterial and venous systems are found in many regions of the body. In some parts of the microcirculation (e.g. in the mesenteries), the capillary circulation can be bypassed by wider thoroughfare channels formed by metarterioles (Fig. 6.15A). These have similarities to both capillaries and the smallest arterioles, and have a discontinuous layer of smooth muscle in their walls. Metarterioles can deliver blood directly to venules or to a capillary bed, according to local demand and conditions. When functional demand is low, blood flow is largely limited to the bypass channel. Periodic opening and closing of different arterioles irrigates different parts of the capillary network. The number of capillaries in individual microvascular units and the size of their mesh determine the degree of vascularity of a tissue; the smallest meshes occur in the lungs and the choroid of the eye.

Arteriovenous anastomoses (Fig. 6.15B) are direct connections between smaller arteries and veins. Connecting vessels may be straight or coiled, and often possess a thick muscular tunic. Under sympathetic control, the vessel is able to close completely, diverting blood into the capillary bed. When patent, the vessel carries blood from artery to vein, partially or completely excluding the capillary bed from the circulation. Simple arteriovenous anastomoses are widespread and occur notably in the skin of the nose, lips and ears, nasal and gastrointestinal mucosae, erectile tissue, tongue, thyroid gland and sympathetic ganglia. In the newborn child, there are few arteriovenous anastomoses but they develop rapidly during the early years. In old age they atrophy, sclerose and diminish in number. These factors may contribute to the less efficient temperature regulation that occurs at the two extremes of age.

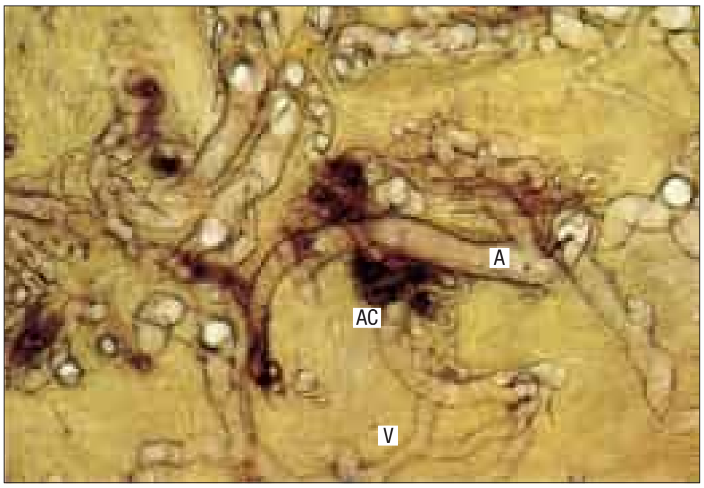

Fig. 6.16 A digital arteriovenous anastomosis, prepared by intravascular perfusion of stain in a full-thickness specimen of skin, followed by clearance. The heavily stained, thick-walled, tortuous anastomotic channels (AC) contrast with the central arterial stem (A) and the thin-walled venous (V) outflow channels. (Courtesy of the late Professor RT Grant, GKT School of Medicine, London.)

In the skin of the hands and feet, especially in digital pads and nail beds (see Fig. 7.18), anastomoses form a large number of small units termed glomera. Each glomus organ has one or more afferent arteries, stemming from branches of cutaneous arteries that approach the surface. The afferent artery gives off a number of fine periglomeral branches and then immediately enlarges, makes a sinuous curve, and narrows again into a short, funnel-shaped vein that opens at right angles into a collecting vein (Fig. 6.16).

Arterial anastomoses

Arteries can be joined to each other by an anastomosis, so that one can supply the territory of the other. An end-to-end anastomosis occurs when two arteries communicate directly, e.g. the uterine and ovarian arteries, the right and left gastro-epiploic arteries, the ulnar artery and the superficial palmar branch of the radial artery. Anastomosis by convergence occurs when two arteries converge and merge, as happens when the vertebral arteries form the basilar artery at the base of the brain. A transverse anastomosis occurs when a short arterial vessel links two large arteries transversely, e.g. the anastomoses between the two anterior cerebral arteries; the posterior tibial artery and the fibular artery; and the radial and ulnar arteries at the wrist.

The angiosome concept and vascular territories

The blood supply to the skin and underlying tissues has been mapped using ink injection studies, corrosion casts, dissection, perforator mapping and two- and three-dimensional radiographic analyses of fresh cadavers and isolated limbs injected with solutions containing lead oxide.

Findings from these different studies all support the clinically important concept of the angiosome, a three-dimensional block of tissue supplied by a source artery, its perforating branches and their accompanying veins (Taylor and Palmer 1987, Levy et al 2003, Taylor 2003, Pan and Taylor 2009). Angiosomes form a complex three-dimensional jigsaw puzzle; some pieces have a predominantly cutaneous component while others are predominantly muscular. Each consists of arteriosomes and venosomes linked to neighbouring angiosomes by either similar calibre (true) or reduced calibre (choke) anastomoses. Anastomoses between adjacent angiosomes may occur within the skin or within muscle. Some muscles are supplied by a single artery and its accompanying veins and therefore lie within one angiosome, while other muscles are supplied by more than one vessel and therefore cross more than one angiosome. A detailed knowledge of anatomical vascular territories and spatial angiosome architecture is essential in designing, evaluating and raising axial flaps based on vascular connections in plastic and reconstructive surgery.

MICROSTRUCTURE OF BLOOD VESSELS

Intima

The intimal lining of blood vessels consists of an endothelium, and a variable amount of subendothelial connective tissue, depending on the vessel.

Endothelium

The endothelium is a monolayer of flattened polygonal cells that extends continuously over the luminal surface of the entire vascular tree (see Fig. 6.10; **Fig. 6.17**). Its structure varies in different regions of the vascular bed.

The endothelium is a key component of the vessel wall and subserves several major physiological roles. Endothelial cells are in contact with the blood stream and thus influence blood flow. They regulate the diffusion of substances and migration of cells out of and into the circulating blood. In the brain, endothelial cells of small vessels actively transport substances, e.g. glucose, into the brain parenchyma. Endothelial cells play an important role in haemostasis because they produce von Willebrand factor, which promotes platelet adhesion (see below); secrete prostacyclin and thrombomodulin, which limits clot formation; and promote fibrinolysis by secreting tissue plasminogen activator. They have selective phagocytic activity and are able to extract substances from the blood. For example, the endothelium of pulmonary vessels removes and inactivates several polypeptides, biogenic amines, bradykinin, prostaglandins and lipids from the circulation, and converts the precursor of angiotensin II to its active form.

Endothelial cells secrete vasoconstrictor (thromboxane) and vasodilator (prostacyclin) prostaglandins, nitric oxide (NO) and endothelin (a vasoconstrictor). They respond to stretch (e.g. increased pressure) and the shear effect of blood flow via stretch-sensitive ion channels in the cell membrane. Endothelial cells synthesize components of the basal lamina. They proliferate to provide new cells during growth of blood vessels, to replace damaged endothelium, and to provide solid cords of cells that develop into new blood vessels (angiogenesis). Angiogenesis, which may be stimulated by endothelial production of autocrine growth factors in response to local hypoxia, is important in wound healing and in the growth of tumours. Endothelial cells are also active participants in, and regulators of, inflammatory processes (Pober and Sessa 2007).

Fig. 6.17 A scanning electron micrograph of the luminal surface of the basilar artery. The tightly packed endothelial cells are elongated in the direction of blood flow. (Courtesy of Masoud Alian, University College, London.)

Although endothelial cells are thin, they extend over a relatively large surface area. They are thickest at the nucleus, where they can reach 2–3 µm, causing a slight bulge into the lumen (see Fig. 6.10). Elsewhere, they are thin and laminar, and are often as little as 0.2 µm thick in capillaries. They are generally elongated in the direction of blood flow, especially in arteries (see Fig. 6.17).

Except in sinusoids, endothelial cells usually adhere to each other at their edges, so there is no discontinuity in the lining of the lumen. They adhere to adjacent cells through the junctional complex, an area of apposition where adherent and tight junctions are found. They also communicate via gap junctions, which are most marked in continuous capillaries. Cell contacts and myoendothelial gap junctions between endothelial and smooth muscle cells are common in small resistance arteries and arterioles, where the separation between endothelium and media is reduced and the inner elastic lamina is either very thin or absent. Myoendothelial gap junctions play an important role in the integration and regulation of vascular function (Figueroa and Duling 2009).

Transcytotic (pinocytotic) vesicles (see Fig. 6.11) are present in all endothelial cells but are particularly numerous in exchange vessels; they include caveolae (see Fig. 6.4) typical of smooth muscle cells. They shuttle small amounts of interstitial fluid or plasma across the endothelial cytoplasm and thus facilitate bulk exchange of nutrients and metabolites between these compartments. They are normally the only means by which protein can cross the endothelium. Endothelial cells secrete many factors but they do not have the morphological characteristics of secretory cells.

The Wiebel–Palade body is characteristic of endothelial cells. It is an elongated cytoplasmic vesicle, 0.2 × 2–3 µm in length, which contains regularly spaced tubular structures parallel to its long axis. Wiebel–Palade bodies play a role in both inflammation and haemostasis because they store the adhesion molecule P-selectin (see below) and von Willebrand factor, which mediates platelet adhesion to the extracellular matrix after vascular injury.

Von Willebrand factor

Von Willebrand factor (vWF) is a very large, multimeric glycoprotein consisting of at least 80 monomers of 250 kDa each, responsible for the characteristic appearance of Wiebel–Palade bodies. Its function is to bind to other proteins and provide a bridge between them.

Endothelial cell–leukocyte interactions

The luminal surface of endothelial cells does not normally support the adherence of leukocytes or platelets. However, many functions of human vascular endothelial cells are dynamic rather than fixed. Activated endothelial cells and the characteristic endothelium of high endothelial venules (HEVs) of lymphoid tissues are sites of leukocyte attachment and diapedesis (see below).

On release from endothelial cells, vWF binds to collagen in the subendothelial matrix; when this is exposed following vascular damage, vWF facilitates platelet adhesion by binding to platelet glycoprotein receptors. This is particularly efficient in high-flow/shear-stress conditions, when it also causes platelet activation. Although the major source of vWF is probably endothelial cells, it is also produced by megakaryocytes and stored in platelet α-granules. vWF is present in plasma, where it binds and stabilizes factor VIII, a clotting protein secreted into the blood stream by hepatocytes. Hereditary deficiency or defective function of vWF causes defective haemostasis and a tendency to bleed (von Willebrand disease) (reviewed in Sadler (2005)).

HEVs (Fig. 6.18) are located within the T-cell domains, between and around lymphoid follicles in all secondary lymphoid organs and tissues except the spleen. They are specialized venules of 7–30 µm diameter, which possess a conspicuous cuboidal endothelial lining. The luminal aspect of HEVs shows a cobblestone appearance. The endothelial cells rest on a basal lamina and are supported by pericytes and a small amount of connective tissue (Fig. 6.19). They are linked by discontinuous adhesive junctions at their apical and basal aspects; the junctions are circumnavigated by migrating lymphocytes. Ultrastructurally, the endothelial cells have the characteristics of metabolically active secretory cells. Thus they contain large, rounded, euchromatic nuclei with one or two nucleoli, prominent Golgi complexes, many mitochondria, ribosomes and pinocytotic vesicles. Typically, they also possess Wiebel–Palade bodies (see above).

Many of the adhesion molecules that mediate interactions between blood leukocytes and HEVs or cytokine-activated endothelium have been identified. They can be divided into three general families: selectins, integrins and the immunoglobulin supergene family. Selectins and integrins are expressed on leukocytes and mediate adhesion of circulating cells to the endothelium, which expresses selectins and members of the immunoglobulin supergene family. Regulated expression of these molecules by both cell types provides the means by which leukocytes recognize the vessel wall (leukocyte homing antigens and vascular addressins), adhere to it and subsequently leave the circulation.

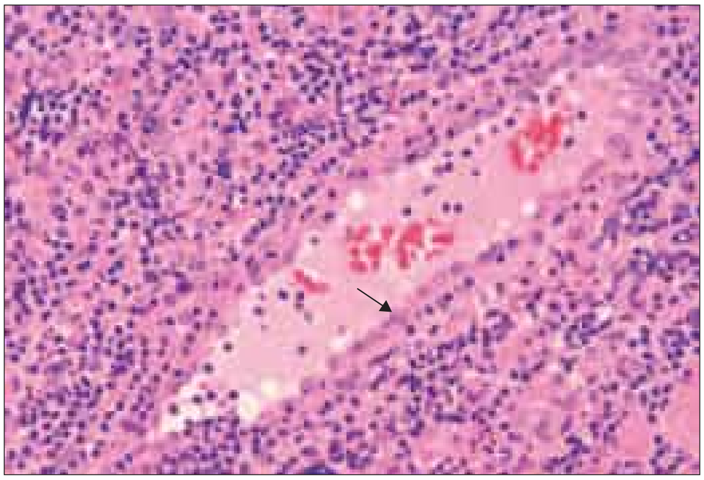

Fig. 6.18 A high endothelial venule in a human lymph node, sectioned longitudinally, lined by cuboidal endothelium (arrow). Erythrocytes and leukocytes (mainly lymphocytes and neutrophils) are seen in the lumen. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

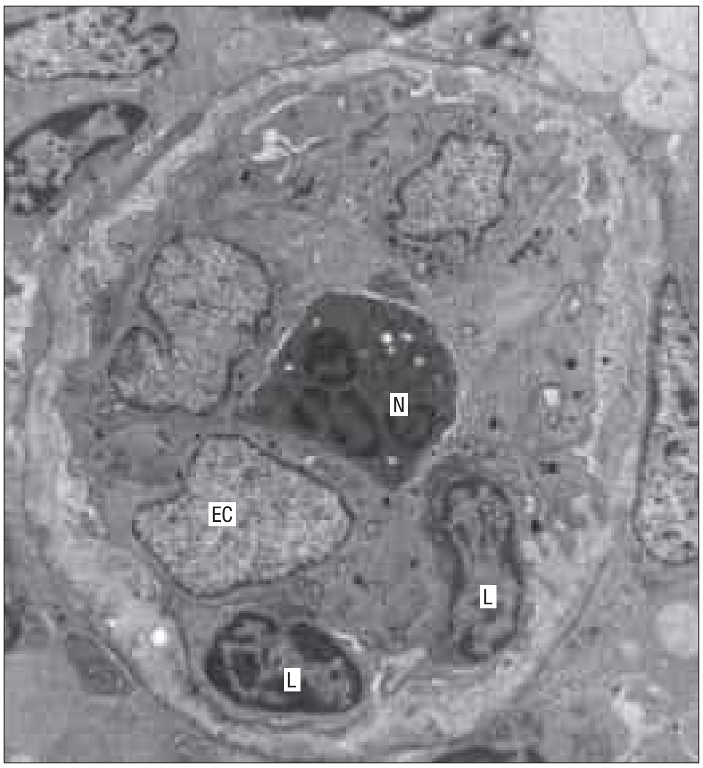

Fig. 6.19 A high endothelial venule in transverse section in a human palatine tonsil. The lumen is completely filled by a neutrophil (N). Cuboidal endothelial cells (EC) line the vessel. Two lymphocytes (L) with heterochromatic nuclei are seen below, in transit within the wall of the vessel. (Courtesy of Dr Marta Perry, Department of Anatomy, St Thomas's Hospital Medical School, London.)

The first step in this cascade is the loose binding or tethering of leukocytes, and this is initiated via L-, P- or E-selectin. This weak, reversible adhesion allows leukocytes to roll along the endothelial surface of a vessel lumen at low velocity, making and breaking contact, and sampling the endothelial cell surfaces. Recognition of chemokines (chemotactic signalling molecules) presented by the endothelium leads to 'inside-out' signalling and conversion of integrins at the leukocyte surface into actively adhesive configurations that bind strongly to their endothelial ligands, resulting in stable arrest. Finally, the leukocyte migrates through the vessel wall (diapedesis), passing either between (paracellular migration) or across (transcellular migration) endothelial cells. Transcellular migration is thought to be the preferred pathway; endothelial transcytotic vesicles (caveolae), intermediate filaments (vimentin) and F-actin are important in the creation of transient transcellular channels through which leukocytes pass. They then cross the basal lamina and migrate into the surrounding tissue by mechanisms that involve CD31 antigen and matrix metalloproteinases (reviewed in Vestweber (2007)).

Cell adhesion molecules

There are three known members of the selectin family of adhesive proteins: L-selectin (also known as lymphocyte homing receptor), E-selectin and P-selectin. L-selectin is expressed on most leukocytes. Endothelial cells of HEVs in lymphoid organs express its oligosaccharide ligand, although other molecules such as mucins may be alternative ligands. L-selectin mediates homing of lymphocytes, especially to peripheral lymph nodes, but also promotes the accumulation of neutrophils and monocytes at sites of inflammation. E-selectin is an inducible adhesion molecule that mediates adhesion of leukocytes to inflammatory cytokine-activated endothelium, and is only transiently expressed on endothelium. P-selectin is rapidly mobilized from Wiebel–Palade bodies, where it is stored, to the endothelial surface after endothelial activation. It binds to ligands expressed on neutrophils, platelets and monocytes, and, like E-selectin, tethers leukocytes to endothelium at sites of inflammation. P-selectin is quickly endocytosed by the endothelial cells and so its expression is short-lived.

The integrins are a large family of molecules that mediate cell-to-cell adhesion as well as interactions of cells with extracellular matrix. Certain β_1 integrin heterodimers are expressed on lymphocytes 2–4 weeks after antigenic stimulation (very late antigens, VLAs) and bind to the extracellular matrix. Additionally, VLA-4, present on resting lymphocytes (expression increases after activation), monocytes and eosinophils, binds to the vascular cell adhesion molecule-1 (VCAM-1), the ligand on activated endothelium. In contrast to β_1 integrins, which many cells express, the expression of β_2 integrins is limited to white blood cells. Although the leukocyte integrins are not constitutively adhesive, they become highly adhesive after cell activation and therefore play a key role in the events required for cell migration. The endothelial ligands for one such β_2 integrin are the intercellular adhesion molecules-1 and 2 (ICAM-1 and ICAM-2), which belong to the immunoglobulin superfamily.

Three members of the large immunoglobulin superfamily of proteins are involved in leukocyte–endothelial adhesion, providing integrin counter-receptors on the endothelial cell membrane. ICAM-1 and ICAM-2 are constitutively expressed but upregulated by inflammatory cytokines. VCAM-1 is absent from resting endothelium but is induced by cytokines on activated endothelium and promotes extravasation of lymphocytes at sites of inflammation.

Subendothelial connective tissue

The subendothelial connective tissue, also termed the lamina propria, is a thin but variable layer. It is largely absent in the smallest vessels, where the endothelium is supported instead by pericytes (see Fig. 6.21). It contains a typical fibrocollagenous extracellular matrix, a few fibroblasts and occasional smooth muscle cells. Endothelial von Willebrand factor concentrates in this layer and participates in haemostasis and platelet adhesion when the overlying endothelium is damaged.

Media

The media consists chiefly of concentric layers of circumferentially or helically arranged smooth muscle cells with variable amounts of elastin and collagen.

Smooth muscle

Smooth muscle forms most of the media of arteries (see Fig. 6.7) and arterioles. A thinner layer of smooth muscle is also found in venules and veins; small segments of the pulmonary veins nearest to the heart contain striated cardiac muscle. Contraction of the smooth muscle in arteries and arterioles reduces the calibre of the vessel lumen, reducing blood flow through the vessel. This is particularly effective in small resistance vessels, where the wall is thick relative to the diameter of the vessel. Smooth muscle activation also increases the rigidity of the vessel wall, reducing its compliance. In arteries this affects propagation of the pulse, whereas in veins it effectively reduces their capacity.

The smooth muscle cells synthesize and secrete elastin, collagen and other extracellular components of the media, which bear directly on the mechanical properties of the vessels. The mechanics of the musculature of the media are complex. Distensibility, strength, self-support, elasticity, rigidity, concentric constriction, etc. are interrelated functions and are finely balanced in the different regions of the vascular bed.

In large arteries, where the blood pressure is high, the muscle cells are shorter (60–200 μm) and smaller in volume than in visceral muscle. In arterioles and veins, smooth muscle cells more closely resemble those from the viscera. The cells are packed with myofilaments and other elements of the cytoskeleton, including intermediate filaments. Vascular muscle cells contain intermediate filaments of either vimentin alone or both vimentin and desmin, whereas the intermediate filaments of visceral smooth muscle are exclusively of desmin. Intercellular junctions are mainly of the adhesive (adherens) type, coupling cells mechanically. Gap junctions couple cells electrically and allow passage of small signalling molecules. Junctions between muscle cells and the connective tissue matrix are particularly numerous, especially in arteries.

The muscle cells of the arterial media can be regarded as multifunctional mesenchymal cells. After damage to the endothelium, muscle cells migrate into the intima and proliferate, forming bundles of longitudinally orientated cells that reform the layer. In certain pathological conditions, muscle cells (and macrophages) undergo fatty degeneration and participate in the formation of atheromatous plaques.

Collagen and elastin

Components of the extracellular matrix (see Ch. 2) are major constituents of vessel walls; in large arteries and veins they make up more than half of the mass of the wall, mainly in the form of collagen and elastin. Other fibrous components such as fibronectin, and amorphous proteoglycans and glycosaminoglycans, are present in the interstitial space.

Elastin is found in all arteries and veins and is especially abundant in elastic arteries (see Fig. 6.6). Individual elastic fibres (0.1–1.0 μm in diameter) anastomose with each other to form net-like structures, which extend predominantly in a circumferential direction. More extensive fusion produces lamellae of elastic material, which, though usually perforated and thus incomplete, separate the layers of muscle cells. The internal elastic lamina is a conspicuous elastic lamella in arteries, between intima and media, which allows the vessel to recoil after distension. When the intraluminal pressure falls below physiological limits (post mortem), it is compressed and coils up into a regular corrugated shape (**Fig. 6.20**; see Figs 6.7, 6.10), reducing but not obliterating the lumen; the profile of the artery remains circular. Fenestrations in the elastic lamina, which may also be split in thickness, allow materials to diffuse between intima and media. An outer elastic lamina, similar in appearance to, but markedly less well developed and less compact than, the internal elastic lamina, lies at the outer aspect of the media at its boundary with the adventitia (see Fig. 6.7). These laminae are less evident in elastic arteries, where elastic fibres occupy much of the media (see Fig. 6.6).

Collagen fibrils are found in all three vessel layers. Type III collagen (reticulin) occupies much of the interstitial space between the muscle cells of the media and is also found in the intima. Collagen is abundant in the adventitia, where type I collagen fibres form large bundles that increase in size from the junction with the media to the outer limit of the vessel wall. In veins, collagen is the main component of the vessel wall and accounts for more than half its mass.

In general terms, collagen and elastic fibres in the media run parallel to, or at a small angle to, the axes of the muscle cells, and they are therefore arranged mainly circumferentially. In contrast, the predominant arrangement of collagen fibres in the adventitia is longitudinal. This arrangement imposes constraints on length change in large vessels under pressure, e.g. in large arteries, in which the radial distension under the effect of the pulse far exceeds the longitudinal distension. The outer sheath of type I collagen in the adventitia therefore has a structurally supportive role. The more delicate type III collagen network of the media provides attachment to the muscle cells and its role is to transmit force around the circumference of the vessel. In a distended vessel, the elastic fibres store energy and, by recoiling, help to restore the resting length and calibre.

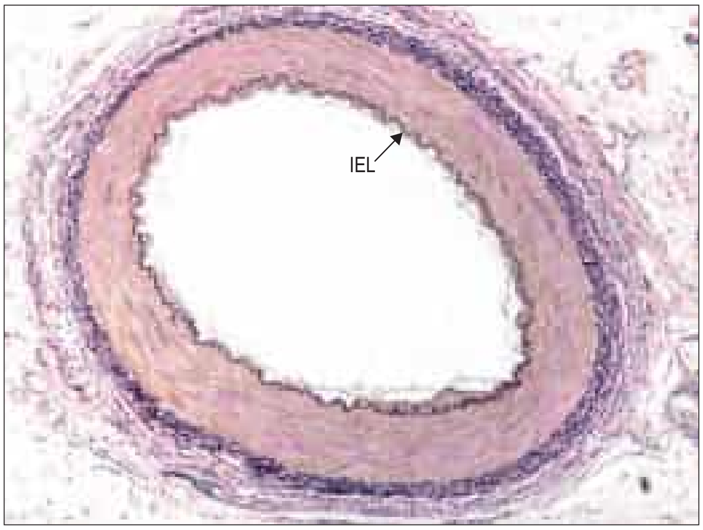

Fig. 6.20 The wall of a human muscular artery, stained for elastin (black), showing the internal elastic lamina (IEL). Fine, incomplete elastic lamellae are interspersed between smooth muscle cells of the tunica media. The endothelium is partly detached in this postmortem specimen. van Gieson stain.

The extracellular material of the media, including collagen and elastin, is produced by the muscle cells. Its turnover is slow compared to that in other tissues. In the adventitia, collagen is synthesized and secreted by fibroblasts, as in other connective tissues. During postnatal development, while vessels increase in diameter and wall thickness, there is an increase in elastin and collagen content. Subsequent changes in vessel structure, seen during ageing, include an increase in the ratio of collagen to elastin, with a reduction in vessel elasticity.

Adventitia

The adventitia is formed of general connective tissue, varying in the thickness and density of its collagen fibre bundles.

Vasa vasorum

In smaller vessels, the nourishment of the tissues of the vessel wall is provided by diffusion from the blood circulating in the vessel itself. The wall thickness at which simple diffusion from the lumen becomes insufficient is 1 mm. Large vessels have their own vascular supply within the adventitia (see Fig. 6.8), in the form of a network of small vessels, the vasa vasorum. These originate from, and drain into, peripheral branches of the vessel they supply. They ramify within the adventitia and, in the largest of arteries, penetrate the outermost part of the media. The vasa vasorum of the pulmonary artery arise from adjacent systemic arteries. The larger veins are also supplied by vasa vasorum but these may penetrate the wall more deeply, perhaps because of the lower oxygen tension.

Nervi vasorum

Blood vessels are innervated by efferent autonomic fibres that regulate the state of contraction of the musculature (muscular tone) and thus the diameter of the vessels, particularly the resistance arteries and arterioles. Perivascular nerves branch and anastomose within the adventitia of an artery, forming a meshwork around it. Nerves are occasionally found within the outermost layers of the media in some of the large muscular arteries.

Nervi vasorum are small bundles of axons, which are almost invariably unmyelinated and typically varicose. Most are postganglionic axons derived from neurones in sympathetic ganglia. The density of innervation varies in different vessels and in different areas of the body; it is usually sparser in veins and larger lymphatic vessels. Large veins with a pronounced muscle layer, such as the hepatic portal vein, are well innervated. Some vessels in the brain may be innervated by intrinsic cerebral neurones, although neural control of brain vessels is of minor importance compared with metabolic and autoregulation (local response to stretch stimuli).

The control of vascular smooth muscle is complex. Vasoconstrictor adrenergic fibres release noradrenaline (norepinephrine), which acts on α-adrenergic receptors in the muscle cell membrane. In addition, circulating hormones and factors such as nitric oxide, prostaglandins and endothelin, which are released from endothelial cells, exert a powerful effect on the muscle cells. Neurotransmitters reach the muscle from the adventitial surface of the media, whereas hormonal and endothelial factors diffuse from the intimal surface. In a few tissues, sympathetic cholinergic fibres inhibit smooth muscle contraction and induce vasodilation. Vascular smooth muscle exhibits endogenous (myogenic) activity in response to stretch and shear.

Most arteries are accompanied by nerves that travel in parallel with them to the peripheral organs that they supply. However, these paravascular nerves are quite independent and do not innervate the vessels they accompany.

Pericytes

Pericytes are present at the outer surface of capillaries and the smallest venules (postcapillary venules), where an adventitia is absent and there are no muscle cells. They are elongated cells, whose long cytoplasmic processes are wrapped around the endothelium. Pericytes are scattered in a discontinuous layer around the outer circumference of capillaries. They are generally absent from fenestrated capillaries but form a more continuous layer around postcapillary venules (**Fig. 6.21**). They are gradually replaced by smooth muscle cells as vessels converge and increase in diameter.

Pericytes are enclosed by their own basal lamina, which merges in places with that of the endothelium. Most display areas of close apposition with endothelial cells, and occasionally form adherens junctions where their basal laminae are absent. Pericyte cytoplasm contains actin, myosin, tropomyosin and desmin, which suggests that they are capable of contractile activity. They also have the potential to act as mesenchymal stem cells, and participate in repair processes by proliferating and giving rise to new blood vessel and connective tissue cells. Pericytes, or closely related cells, may be the source of myofibroblasts that contribute to fibrosis in disease processes (reviewed in Duffield (2012)).

Cerebral vessels

Major branches of cerebral arteries that lie in the subarachnoid space over the surface of the brain have a thin outer coating of meningeal cells, usually one layer thick, where adjacent meningeal cells are joined by desmosomes and gap junctions. These arteries have a smooth muscle media and a distinct elastic lamina. Veins on the surface of the brain have very thin walls, and the smooth muscle layers in the wall are often discontinuous. They are coated externally by a monolayer of meningeal cells.

As arteries enter the subpial space and penetrate the brain, they lose their elastic laminae, and consequently the cerebral cortex and white matter typically contain only arterioles, venules and capillaries. The exceptions are the large penetrating vessels in the basal ganglia, where many arteries retain their elastic laminae and thick smooth muscle media. Enlarged perivascular spaces form around these large arteries in ageing individuals. Arterioles and venules in the cortex and white matter can be distinguished from each other because arterioles are surrounded by a smooth muscle coat, whereas veins and venules have larger lumina and thinner walls.

Cerebral capillaries are the site of the blood–brain barrier. They are lined by endothelial cells, which are joined by tight junctions. The endothelial cytoplasm contains a few pinocytotic vesicles. The cells are surrounded by a basal lamina (see Fig. 3.12); at points of contact with perivascular astrocytes, the intervening basal lamina is formed by fusion of the endothelial and glial basal laminae. Pericytes, completely surrounded by basal lamina, are present around capillaries. Perivascular macrophages are attached to the outer walls of capillaries and to other vessels; they are phenotypically distinct from parenchymal microglia, which are also of monocytic origin. A thin layer of meningeal cells derived from the pia mater surrounds arterioles but disappears at the level of capillaries. For further descriptions of cerebral vessels, see Chapter 19.

LYMPHATIC VESSELS

Lymphatic capillaries form wide-meshed plexuses in the extracellular matrices of most tissues. They begin as dilated, blind-ended tubes with larger diameters and less regular cross-sectional appearances than those of blood capillaries. Their basal laminae are incomplete or absent and they lack associated pericytes. The smaller lymphatic vessels are lined by endothelial cells, which have numerous transcytotic vesicles within their cytoplasm and so resemble blood capillaries. However, unlike capillaries, their endothelium is generally quite permeable to much larger molecules: they are readily permeable to large colloidal proteins and particulate material such as cell debris and microorganisms, and also to cells. Permeability is facilitated by gaps between the endothelial cells, which lack tight junctions (discontinuous endothelium), and by pinocytosis. The lymphatic system provides an important transport pathway for leukocytes and defence against infection (reviewed in Saharinen et al (2004)).

Lymph is formed from interstitial fluid, which is derived from blood plasma via filtration in the microcirculation. Much of the filtered fluid is reabsorbed by the time the blood leaves the venules, but about 15% or 8 litres per day enters the lymphatics. Lymphatic vessels take up this residual fluid by passive diffusion and the transient negative pressures in their lumina, which are generated intrinsically by contractile activity of smooth muscle in the largest lymphatic vessel walls, and extrinsically by compression of the lymph vessels as a result of contraction of adjacent muscle or arterial pulsation. The unidirectional flow of lymph is maintained by the presence of valves in the larger vessels (**Fig. 6.22**). Lymphatic capillaries are prevented from collapsing by anchoring filaments, which tether their walls to surrounding connective tissue structures and exert radial traction.

In most tissues, lymph is clear and colourless; in the lymphatic capillaries it has an identical composition to interstitial fluid. In contrast, lymph from the small intestine is dense and milky, reflecting the presence of lipid droplets (chylomicrons) derived from fat absorbed by the mucosal epithelium. The terminal lymphatic vessels in the mucosa of the small intestine are known as lacteals and the lymph as chyle. Lymphatic capillaries are not ubiquitous: they are not present in cornea, cartilage, thymus, the central or peripheral nervous system or bone marrow, and there are very few in the endomysium of skeletal muscles.

Lymphatic capillaries join into larger vessels that pass to local lymph nodes. Typically, lymph percolates through a series of nodes before

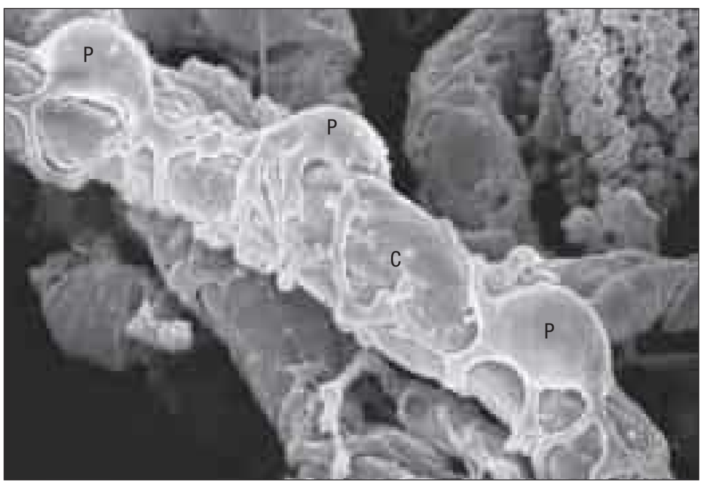

Fig. 6.21 A scanning electron micrograph of capillary (C) and pericytes (P) supporting the vessel wall. (Courtesy of T. Fujiwara and Y. Uehara, Department of Anatomy, Ehime University School of Medicine, Japan.)

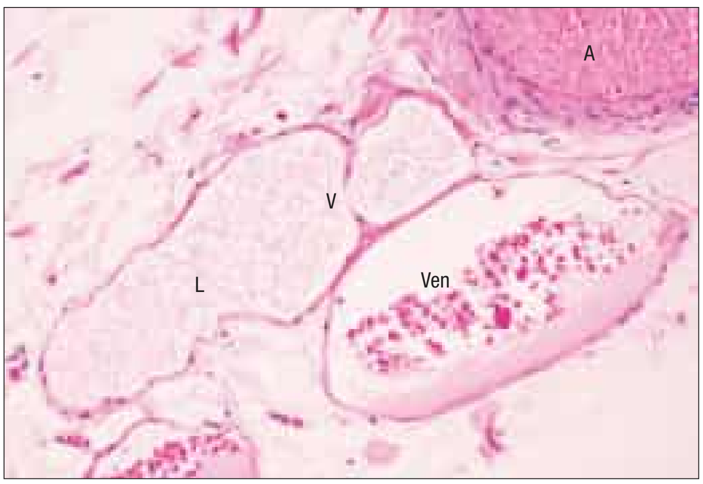

Fig. 6.22 A valve (V) in a lymphatic vessel (L), accompanying a small venule (Ven) and arteriole (A) in human connective tissue. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

reaching a major collecting duct. There are exceptions to this arrangement: the lymph vessels of the thyroid gland and oesophagus, and of the coronary and triangular ligaments of the liver, all drain directly to the thoracic duct without passing through lymph nodes. In the larger vessels, a thin external connective tissue coat supports the endothelium. The largest lymphatic vessels (200 µm) have three layers, like small veins, although their lumen is considerably larger than that of veins with a similar wall thickness. The tunica media contains smooth muscle cells, mostly arranged circumferentially. Elastic fibres are sparse in the tunica intima, but form an external elastic lamina in the tunica adventitia.

The larger lymphatic vessels differ from small veins in having many more valves (see Fig. 6.22). The valves are semilunar, generally paired and composed of an extension of the intima. Their edges point in the direction of the current, and the vessel wall downstream is expanded into a sinus, which gives the vessels a beaded appearance when they are distended. Valves are important in preventing the backflow of lymph.

Deep lymphatic vessels usually accompany arteries or veins, and almost all reach either the thoracic duct or the right lymphatic duct, which usually joins the left or right brachiocephalic veins respectively at the root of the neck.

The thoracic duct is structurally similar to a medium-sized vein but the smooth muscle in its tunica media is more prominent. Most lymphatic vessels anastomose freely. Larger vessels have their own plexiform vasa vasorum and accompanying nerve fibres. If their walls are acutely infected (lymphangitis) this vascular plexus becomes congested, marking the paths of superficial vessels by red lines that are visible through the skin and tender to the touch.

Lymphatic vessels repair easily. New vessels form readily after damage, beginning as solid cellular sprouts from the endothelial cells of persisting vessels that subsequently become canalized.

Lymphoedema

Insufficient lymphatic drainage leads to accumulation of fluid in the tissues (oedema) and swelling, typically in the limbs.

CARDIAC MUSCLE

In cardiac muscle, as in skeletal muscle, the contractile proteins are organized structurally into sarcomeres that are aligned in register across the fibres, producing fine cross-striations that are visible in the light microscope. Both types of muscle contain the same contractile proteins (although commonly different isoforms), which are assembled in a similar way. The molecular basis for contraction, but not its regulation, is the same. Elevation of cytosolic calcium triggers contraction, corresponding to cardiac systole and the pumping phase of the heart cycle. Removal of cytosolic calcium induces relaxation, corresponding to diastole and cardiac filling. Despite the similarities, there are major functional, morphological and developmental differences between cardiac and skeletal muscle.

MICROSTRUCTURE OF CARDIAC MUSCLE

The myocardium, the muscular component of the heart, constitutes the bulk of its tissues. It consists predominantly of cardiac muscle cells, which are usually 120 µm long and 20–30 µm in diameter in a normal adult. Each cell has one or two large nuclei occupying the central part of the cell, whereas skeletal muscle has multiple, peripherally placed nuclei. The cells are branched at their ends, and the branches of adjacent cells are so tightly associated that their appearance under light microscopy is of a network of branching and anastomosing fibres (**Fig. 6.23**). Cells are bound together by elaborate junctional complexes, the intercalated discs (**Fig. 6.24**; see Figs 6.23, 6.26).

Fine fibrocollagenous connective tissue is found between cardiac muscle fibres. Although this is equivalent to the endomysium of skeletal muscle, it is less regularly organized because of the complex three-dimensional geometry imposed by the branching cardiac cells. Numerous capillaries and some nerve fibres are found within this layer. Coarser connective tissue, equivalent to the perimysium of skeletal muscle, separates the larger bundles of muscle fibres, and is particularly well developed near the condensations of dense fibrous connective tissue that form the 'skeleton' of the heart. The ventricles of the heart are composed of spiralling layers of fibres that run in different directions. Consequently, microscope sections of ventricular muscle inevitably contain the profiles of cells cut in a variety of orientations. A linear arrangement of cardiac muscle fibres is found only in the papillary muscles and trabeculae carneae.

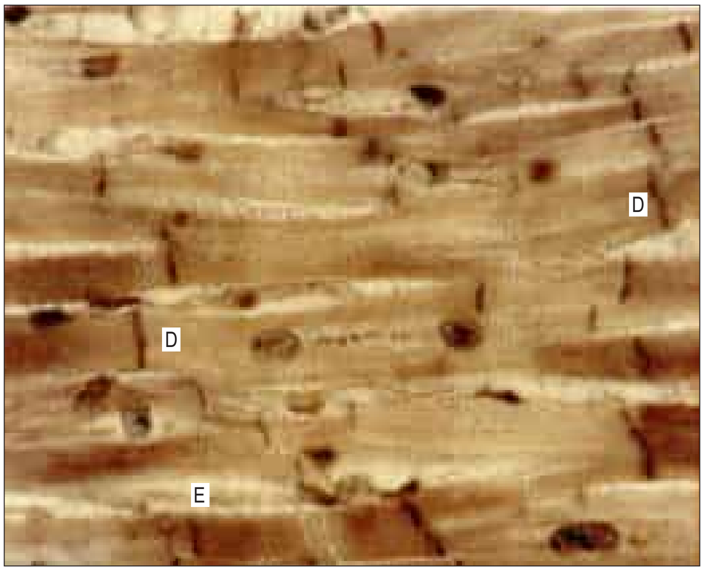

Fig. 6.23 Cardiac muscle fibres in human myocardium, sectioned longitudinally. Cell branching and fine cross-striations are clearly visible and indicate the intracellular organization of sarcomeres. The dark transverse lines are intercalated discs (D). Endomysium (E) contains nuclei of endothelial cells and fibroblasts.

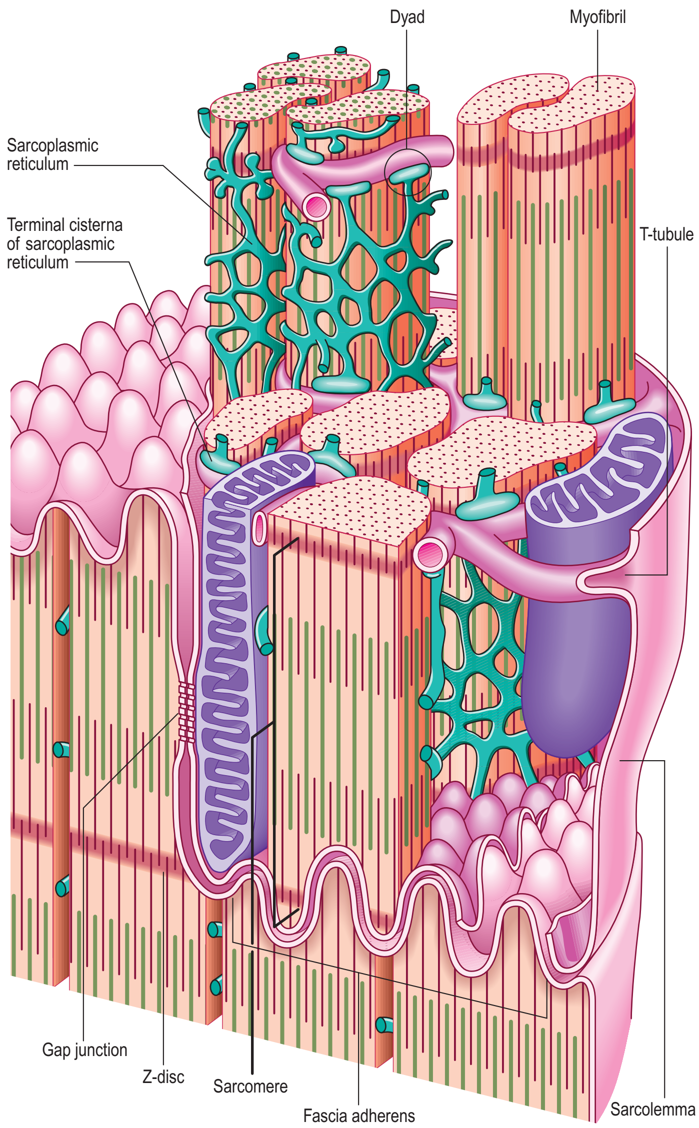

Fig. 6.24 A three-dimensional reconstruction of cardiac muscle cells in the region of an intercalated disc, a junctional complex between neighbouring cells. The interdigitating transverse parts of the intercalated disc form a fascia adherens, with numerous desmosomes; gap junctions are found in the longitudinal parts of the disc. The organization of the transverse tubules and sarcoplasmic reticulum is also shown.

In the developed world, secondary lymphoedema is most commonly the consequence of treatment for cancer, particularly breast cancer, following either surgical removal of lymph nodes and vessels, or their damage by radiation therapy. Tumours can themselves cause lymphoedema if they compress and block lymph vessels. Lymph node infections can also restrict lymph flow. In tropical regions, invasion of the lymphatic vessels by thread-like nematode worms (filariasis) can lead to severe swelling, particularly in the legs and genitals (elephantiasis) (Pfarr et al 2009). Podoconiosis is a non-filarial elephantiasis endemic to tropical highland areas of Africa and elsewhere. It is thought to be due to an abnormal inflammatory reaction to volcanic red clay soils (Fuller 2005). Both filariasis and podoconiosis have a large socioeconomic impact. Primary lymphoedema is a rare inherited condition, most common in females, and may involve abnormal development of lymph nodes or valves in the lymphatic vessels.

Electron micrographs of cardiac muscle cells in longitudinal section show that the myofibrils separate before they pass around the nucleus, leaving a zone that is occupied by organelles, including sarcoplasmic reticulum, Golgi complex, mitochondria, lipid droplets and glycogen (**Fig. 6.25**). At the light microscopic level, these zones appear in longitudinal sections as unstained areas at the poles of each nucleus. They often contain lipofuscin granules, which accumulate there in individuals over the age of 10; the reddish-brown pigment may be visible even in unstained longitudinal sections.

The cross-striations of cardiac muscle are less conspicuous than those of skeletal muscle. This is because the contractile apparatus of cardiac muscle lies within a mitochondria-rich sarcoplasm. The myofibrils are also less well delineated in cardiac muscle; in transverse sections they often fuse into a continuous array of myofilaments, irregularly bounded by mitochondria and longitudinal elements of sarcoplasmic reticulum. The large mitochondria, with their closely spaced cristae, reflect the highly developed oxidative metabolism of cardiac tissue. The proportion of the cell volume occupied by mitochondria (approximately 35%) is even greater in cardiac muscle than it is in slow twitch skeletal muscle fibres. The high demand for oxygen is also reflected in high levels of myoglobin and an exceptionally rich network of capillaries around the fibres.

The force of contraction is transferred through the ends of the cardiac muscle cells via the junctional strength provided by the intercalated discs. As in skeletal muscle, force is also transmitted laterally to the sarcolemma and extracellular matrix via vinculin-containing elements that bridge between the Z-discs of peripheral myofibrils and the plasma membrane. The actin-binding proteins spectrin and dystrophin, important components of the cardiac muscle cell cytoskeleton, associate independently with the sarcolemma to provide mechanical support. Both protein complexes form components of the costamere (Peter et al 2011; Samarel 2005), which, in addition to transmitting force from sarcomeres to the sarcolemma and extracellular matrix, is central to mechanically generated signalling mechanisms.

Atrial muscle cells are smaller than ventricular cells. The cytoplasm near the Golgi complexes at the poles of the nuclei exhibits dense membrane-bound granules that contain the precursor of atrial natriuretic peptide. This hormone promotes loss of sodium and water in the kidneys, reducing blood volume and thereby lowering blood pressure. It is released into the blood in response to stretch of the atrial wall. Atrial natriuretic peptide and aldosterone have antagonistic effects on renal sodium and water handling, via independent mechanisms.

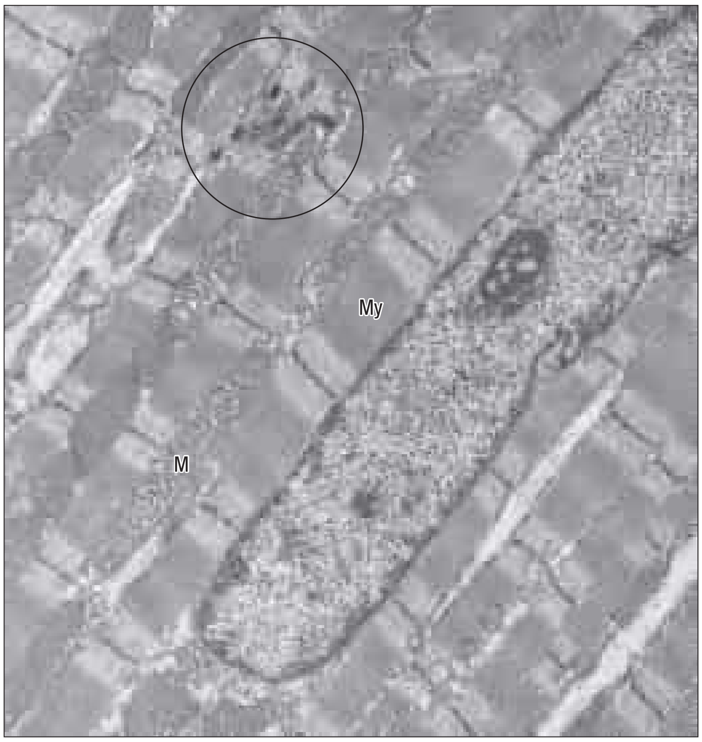

Fig. 6.25 A low-power transmission electron micrograph of cardiac muscle in longitudinal section, including the perinuclear zone of one of the fibres. Note the abundant large mitochondria (M) between myofibrils (My), and an intercalated disc (circled). (Courtesy of Professor Brenda Russell, Department of Physiology and Biophysics, University of Illinois at Chicago.)

The sarcolemma of ventricular cardiac muscle cells invaginates to form T-tubules with a wider lumen than those of skeletal muscle; atrial muscle cells have few or no T-tubules. Unlike skeletal muscle, most T-tubules penetrate the sarcoplasm at the level of the Z-discs (see Fig. 6.24). The T-tubules are interconnected at intervals by longitudinal branches to form a complex network. They serve a similar function in skeletal and cardiac muscle, i.e. to carry the wave of depolarization into the cores of the cells.

The sarcoplasmic reticulum is a membrane-bound tubular plexus that surrounds and defines, sometimes incompletely, the outlines of individual myofibrils. Its main role, as in skeletal muscle, is the storage, release and reaccumulation (sequestration) of calcium ions. The calcium-binding protein calsequestrin allows large amounts of calcium to be stored within the sarcoplasmic reticulum and modulates activity of the calcium release channels (ryanodine receptors) (Györke and Terentyev 2008). The sarcoplasmic reticulum is separated from the T-tubules by a 15 nm gap spanned by structures termed junctional processes, which are thought to be the cytoplasmic part of the calcium release channels; similar processes are found in skeletal muscle at the junctional surface of the terminal cisternae. Sarcoplasmic reticulum bearing junctional processes is termed junctional sarcoplasmic reticulum to distinguish it from free sarcoplasmic reticulum, which forms a longitudinal network. Junctional sarcoplasmic reticulum makes contact with both the T-tubules and the sarcolemma (of which the T-tubules are an extension). Sarcoplasmic reticulum forms small globular extensions (corbular sarcoplasmic reticulum) in the vicinity of the Z-discs, but not in immediate relation to T-tubules or the sarcolemma. Since the junctions between T-tubules and sarcoplasmic reticulum usually involve only one structure of each type, the corresponding profiles in electron micrographs are referred to as dyads, rather than triads as in skeletal muscle.

Intercalated discs

Intercalated discs are unique, complex junctions between cardiac myocytes. In the light microscope they are seen as transverse lines crossing the tracts of cardiac cells (see Fig. 6.23). They may step irregularly within or between adjacent tracts, and may appear to jump to a new position as the plane of focus is altered. Ultrastructurally, they are seen to have transverse and lateral portions (see Fig. 6.24; **Fig. 6.26**). The transverse portions occur wherever myofibrils abut the end of the cell, and each takes the place of the last Z-disc. At this point, the actin filaments of the terminal sarcomere insert into a dense subsarcolemmal matrix that anchors them, together with other cytoplasmic elements such as intermediate filaments, to the plasma membrane. Prominent desmosomes, often with a dense line in the intercellular space, occur at intervals along each transverse portion. This junctional region is

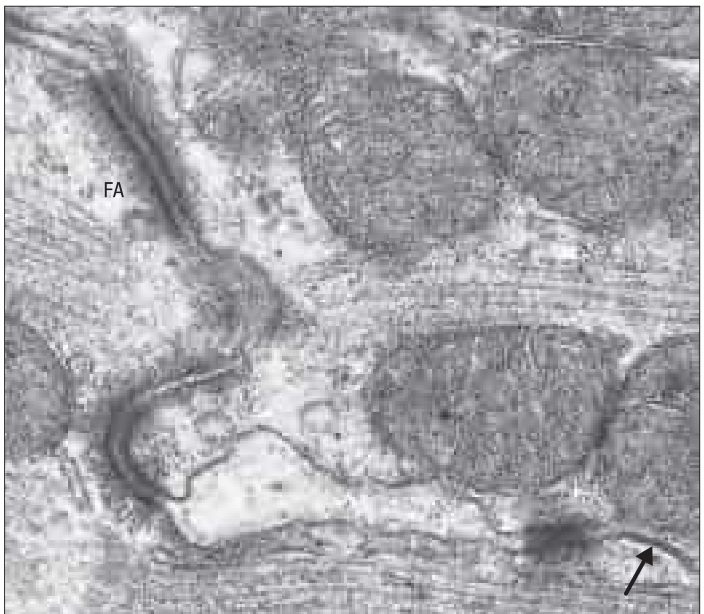

Fig. 6.26 An intercalated disc in cardiac muscle, with several zones of electron-dense fascia adherens (FA) and a gap junction (arrow). (Courtesy of Professor Brenda Russell, Department of Physiology and Biophysics, University of Illinois at Chicago.)

homologous with, and probably similar in composition to, the structure found on the cytoplasmic face of the myotendinous junction, and is a type of fascia adherens junction. It provides firm adhesion between cells, and a route for the transmission of contractile force from one cell to the next.

The lateral portions of an intercalated disc run parallel to the myofilaments and the long axis of the cell, for a distance that corresponds to one or two sarcomeres before turning again to form another transverse portion. They are therefore responsible for the stepwise progression of the intercalated disc, which is seen microscopically. The lateral portions contain gap junctions between adjacent cells (see Fig. 6.26), which provide electrical coupling and so enable the electrical impulse to propagate from one cell to the next, spreading excitation and contraction rapidly along the branching tracts of interconnected cells. The activity of the individual cells of the heart is thus coordinated so that they function as if they were a syncytium.

Contractile protein isoforms of cardiac muscle

As in skeletal muscle, the contractile proteins of cardiac muscle exist in a number of tissue- and stage-specific forms.

Endocardium

The whole inner surface of the heart is covered by a thin layer of cells forming the endocardium, which separates the myocardium from the blood. These cells are very similar to the endothelial cells that line blood vessels, with a similar biological activity and embryological origin. They are now recognized as playing an important signalling role in cardiac muscle function (reviewed in Brutsaert (2003)).

NEUROVASCULAR SUPPLY OF CARDIAC MUSCLE

Vascular supply

The activity of the heart is equivalent to a constant power expenditure of 1.3 watts under basal conditions, and escalates to 3 watts or more during physical exertion. Cardiac muscle cells contain glycogen, which is a reserve during peaks of activity, but the majority of their energy requirement is continuous and supplied only through a highly developed oxidative metabolism, as is evident from the high proportion of the cell volume that is occupied by mitochondria. This metabolism has to be supported by a rich blood supply. Myocardium has a very high perfusion rate of 0.5 ml/min/g of tissue (5 times that of liver and 15 times that of resting skeletal muscle). No cardiac muscle cell is more than 8 μm from a capillary, and vascular channels occupy a high proportion of the total interstitial space. Cardiac muscle is supplied by the coronary vessels. Although there is some variation in the detailed distribution of the arterial branches, the left ventricle, which has the highest workload and largest mass, receives the highest arterial blood flow. Branches run in the myocardium along the coarser aggregations of connective tissue and ramify extensively in the endomysial layer, creating a rich plexus of anastomosing vessels. This plexus includes lymphatic as well as blood capillaries, which is not the case in skeletal muscle.

The high oxygen requirement of the myocardium makes it vulnerable to ischaemic damage arising from atheroma or embolism in the coronary arteries. This vulnerability is exacerbated in the left ventricle because embedded coronary arteries are compressed by contraction of the myocardium, limiting perfusion during systole. Arterial anastomoses, often more than 100 μm in diameter, are found throughout the heart and are an important factor in determining whether an adequate collateral circulation may develop after a coronary occlusion.

Innervation

Although the impulse-generating and conducting system of the heart establishes an endogenous rhythm, the rate and force of contraction are under neural influence. Both divisions of the autonomic nervous system supply unmyelinated postganglionic fibres to the heart. The innervation is derived bilaterally but it is functionally asymmetrical. Activation of the left stellate ganglion (sympathetic) has little effect on heart rate but increases ventricular contractility, whereas activation of the right stellate ganglion influences both rate and contractility. Activation of the right vagus nerve (parasympathetic) slows heart rate mainly through its influence on the pacemaker region, the sinuatrial (SA) node, whereas activation of the left vagus slows propagation of the impulse mainly through its effect on the atrioventricular (AV) node. Vagal activity has little direct effect on ventricular contractility.

Postganglionic sympathetic nerve fibres from the cervical sympathetic ganglia reach the heart via the recurrent cardiac nerves. Parasympathetic fibres in the heart are derived from ganglion cells in the cardiac plexuses and atrial walls that are innervated by preganglionic cardiac branches of the vagus. Adrenergic, cholinergic and peptidergic endings have been demonstrated in the myocardium. Fibres often end close to muscle cells and blood vessels, but junctional specializations are not seen, and a gap of at least 100 nm remains between myocyte and axon. It is probable that neurotransmitters diffuse across this gap to the adjacent cells. Some of the endings represent efferent nerve terminals, while others function as pain receptors, mechanoreceptors or chemoreceptors.

EXCITATION–CONTRACTION COUPLING IN CARDIAC MUSCLE

The molecular interaction between actin and myosin that underlies the generation of force is initiated by elevation of cytosolic calcium in both cardiac and skeletal muscle. However, there are significant differences in both the mechanisms by which this occurs and the physical arrangement and molecular composition of the contractile elements (reviewed in Bers (2002)).

One of the major functional differences between cardiac and skeletal muscle is the way in which contractile force is regulated. Smoothness and gradation of contraction in skeletal muscle depend on recruitment and asynchronous firing of different numbers of motor units. Individual motor units can also build up a contraction through a brief series of re-excitations. In the heart, the entire mass of muscle must be activated almost simultaneously, and mechanical summation by re-excitation is not possible, because the cells are electrically refractory until mechanical relaxation has taken place (see below).

As in skeletal muscle, cardiac muscle contraction is initiated when calcium binds to troponin-C, a component of the regulatory protein complex on the thin filaments. During basal activity of the heart, the amount of calcium bound to troponin-C during each systole induces less than half-maximal activation of the contractile apparatus. Contractile force can therefore be increased by increasing cytosolic calcium and thus the amount bound to troponin-C. This is achieved by controlling the amount of free calcium that is released into the cytosol during systole.

A special feature of the cardiac muscle cell is the long duration of its action potential. This is due to activation of voltage-dependent L-type calcium channels in the sarcolemma by the initial depolarization, through which calcium enters the cell, sustaining the depolarization and so causing a long-lasting electrical plateau of equivalent length to the contraction. However, calcium influx during the plateau only accounts for about 25% of the elevation of cytosolic calcium; the rest is released from the sarcoplasmic reticulum.

Action potentials are conducted into the T-tubules, where they activate L-type channels and calcium influx. This causes a localized increase in free calcium in the narrow space between the T-tubules and junctional sarcoplasmic reticulum, which activates calcium release channels on the latter, causing calcium to flood out into the cytosol. This 'calcium-induced calcium release' is a vital component of cardiac muscle activation and is the principal or only mechanism by which calcium can be released from the sarcoplasmic reticulum in the myocardium. This major difference between cardiac and skeletal muscle is reflected in tissue-specific isoforms of calcium release channels.

Systolic activation is terminated on repolarization of the action potential by reuptake (sequestration) of calcium from the cytosol back into the sarcoplasmic reticulum. This is mediated by a high-affinity sarco-endoplasmic reticulum calcium ATPase (SERCA), which rapidly reduces cytosolic free calcium to resting levels. The activity of this ATPase controls the rate of decay of the calcium transient and is therefore a determinant of the rate of relaxation of the heart. The sarcoplasmic reticulum contains a cardiac form of calsequestrin, a homologue of the proteins found in skeletal and smooth muscle. This calcium-binding protein buffers the free calcium concentration inside the sarcoplasmic reticulum, allowing it to store considerable amounts of total calcium without increasing the gradient against which SERCA must pump. It may also modulate the activity of the calcium release channels (Györke and Terentyev 2008).

Whilst SERCA therefore allows rapid relaxation of the myocardium, the calcium that entered the cell during the action potential must also

The cardiac isoform of α-actin is not identical to the skeletal muscle form and is encoded by a different gene, although the two are so similar as to be functionally interchangeable. Both skeletal and cardiac isoforms of sarcomeric actin are expressed in fetal ventricular muscle. The mRNA for skeletal α-actin increases postnatally and exceeds that of cardiac actin in the adult.

The myosin heavy chain of human cardiac muscle exists in two isoforms, α and β, both of which are present in the fetal heart. The α-form persists as the adult isoform in atrial muscle, whereas the β-form (which is associated with a slower rate of contraction) predominates in ventricular muscle. Interestingly, the β-form of myosin heavy chain in cardiac muscle is identical to the isoform in slow twitch skeletal muscle. This identity between cardiac and slow twitch skeletal protein isoforms is true of several proteins, including ventricular myosin light chains and cardiac troponin-C. Other proteins, such as troponin-I and T, exist in cardiac-specific forms in the adult, although skeletal isoforms are expressed in the fetus and neonate. The appearance of cardiac-specific isoforms of troponin-I and T in the blood, following their release from damaged cardiac cells, is now a standard diagnostic test for myocardial infarction.

be transported out again to prevent calcium overload. This is accomplished primarily by a sodium–calcium exchanger (NCX) in the sarcolemma, and to a much lesser extent by a sarcolemmal calcium ATPase. These act more slowly than SERCA, continuously removing calcium from the cell during the diastolic interval as calcium slowly leaks from the sarcoplasmic reticulum.

The extent of the elevation of cytosolic calcium and therefore force generation during systole is determined by both the size of the initial stimulus (amount of calcium entering through L-type channels) and the amount of calcium stored within the sarcoplasmic reticulum. The greater the amount stored, the more is available for release. This provides an intrinsic mechanism for matching any increase in heart rate with a progressive increase in contractile force. At higher heart rates, more calcium enters per unit time, but as the diastolic interval is reduced there is less time for calcium to be removed from the cell by the sarcolemmal sodium–calcium exchanger. The amount of calcium stored in the sarcoplasmic reticulum therefore increases, so more can be released during systole. Increased heart rate is therefore coupled with an increase in force, a phenomenon called the staircase response.

The most potent physiological means of enhancing cardiac contractility is through the action of β-adrenergic agents, specifically the sympathetic neurotransmitter noradrenaline (norepinephrine) and circulating adrenaline (epinephrine). Activation of β-adrenergic receptors produces cyclic adenosine monophosphate (cAMP), which enhances calcium influx via L-type calcium channels during depolarization, so stimulating greater release from the sarcoplasmic reticulum. In addition, cyclic AMP enhances the activity of SERCA via promoting phosphorylation of an associated protein, phospholamban. This enables a more rapid reduction of the cytosolic calcium pump and accelerated relaxation, which is important when the heart rate is increased. Phosphorylation of troponin-I on the thin filaments also increases the rate of cross-bridge cycling, further accelerating relaxation.

Dysfunction of cardiac muscle: excitation–contraction coupling in disease

In coronary heart disease the myocardium may be inadequately perfused (myocardial ischaemia), leading to energy deficit, loss of contractility and eventually cell death. The process may culminate in irretrievable myocardial damage and progress to heart failure, which is the inability to provide adequate cardiac output for the tissues.

ORIGIN OF CARDIAC MUSCLE

Cardiac myocytes differentiate from the splanchnopleuric coelomic epithelial cells of the pericardium initially subjacent to the endoderm (see Fig. 52.6). As the primitive heart tube is formed, the presumptive myocardial cells start to express genes that encode characteristic myocardial proteins, including myosin, actin, troponin and other components of the contractile apparatus. Myofibrils begin to appear in the developing myoblasts, and the first functional heart beats start soon afterwards.

Committed cardiac myoblasts do not fuse to form multinucleated myotubes, as occurs in skeletal muscle, but remain as single cells coupled physically and electrically through intercellular junctions. Differentiated cardiac myocytes continue to divide during fetal development and withdraw from the cell cycle only after birth. This is markedly different from skeletal muscle development, when differentiation, including the activation of muscle-specific genes, coincides with withdrawal from the cell cycle. During fetal maturation, successive changes in gene expression give rise to the characteristics of fetal, neonatal and adult myocardium, and are responsible for the divergence of the properties of atrial and ventricular muscle cells.

The regulatory mechanisms underlying differentiation of cardiac muscle appear to be distinct from those of skeletal muscle. It is anticipated that counterparts will be found for the transcriptional factors Myf-5, myogenin, MyoD and Myf-6, which are responsible for inducing differentiation of skeletal muscle. It is known that Tbx5 and Nkx2–5 are required for formation of atrial and ventricular myocardium (see Fig. 52.13).

Concurrent with the development of contractile proteins, cardiac myocytes develop numerous specific intracellular vesicles containing substances shown to induce natriuresis and diuresis, and a family of polypeptides generally known as atrial natriuretic peptides. These vesicles develop from the Golgi complex in both atria and ventricles during fetal life but become restricted to atrial muscle in the adult. Atrial natriuretic peptide is measurable when the heart is recognizably four-chambered; almost all cells within the atria are capable of its synthesis.

The impulse-generating (AV node) and conducting systems of the heart (AV node; bundle of His, Purkinje fibres) are formed from cardiac myocytes that differ in their morphology from the working myocardium of the heart chambers. Cells of the SA and AV nodes tend to be smaller with slow cell-to-cell electrical conduction, whilst those of the bundle of His and Purkinje fibres are larger in diameter with very rapid electrical conduction.

CARDIAC PLASTICITY AND REGENERATION OF CARDIAC MUSCLE

The heart exhibits a significant degree of plasticity. Physiological and pathophysiological stimuli can initiate cardiac remodelling (reviewed in Hill and Olson (2008)). For example, increased loads as a result of exercise training or pregnancy induce physiological hypertrophy of the heart, where the ventricular walls thicken due to an increase in the size (but not number) of cardiac muscle cells. Conversely, hypertrophy resulting from cardiac damage, high blood pressure and/or extended activation of neurohumoral compensatory mechanisms (e.g. sympathetic stimulation, production of angiotensin II and aldosterone) can be maladaptive, increasing the risk and rate of progression of heart failure and sudden death due to arrhythmias.

In skeletal muscle, a population of precursor cells (satellite cells) is retained in adult life, constituting a pool of myoblasts capable of dividing, fusing with existing muscle fibres and initiating regeneration after damage. It was thought that cardiac muscle lacked an equivalent population of cells and was therefore incapable of regeneration. There is some evidence that the heart does contain endogenous progenitor cells capable of regenerating myocardial function (Oh et al 2004), although to a very limited extent compared to skeletal muscle and insufficient to restore heart function after ischaemic or other injury. It has been suggested that there is a slow but continuous renewal of cardiac muscle cells throughout life even in healthy individuals, with a turnover of between 0.4 and 1% per year (Bergmann et al 2009).

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The body responds to this challenge with a variety of neurohumoral mechanisms, including increased sympathetic stimulation to the heart, increasing heart rate and contractility via noradrenaline (norepinephrine; see above). Treatment includes administration of positive inotropic agents (drugs that increase contractility) such as β-adrenergic agonists and phosphodiesterase inhibitors that prevent the breakdown of cyclic AMP. The classical positive inotropic drug is digitalis (digoxin), which at therapeutic doses causes partial inhibition of the sodium pump (sodium–potassium ATPase) that maintains the ionic gradient across the sarcolemma. This leads to a reduced sodium concentration gradient, impairing operation of the sarcolemmal sodium–calcium exchanger. Less calcium is therefore extruded from the cell, so more is left in the sarcoplasmic reticulum for release during systole, increasing force. However, positive inotropes are now only used in the short term because, if used chronically, they can hasten progression of the disease. The following explains why this is the case.

An energy deficit has far-reaching consequences beyond the inhibition of actin and myosin interactions. ATP is vital for many cellular processes: in particular, the activities of the sodium pump and SERCA. Impaired operation of the sodium pump leads to reduced extrusion of calcium (see above). Inhibition of SERCA impairs calcium reuptake into the sarcoplasmic reticulum and relaxation at the end of systole. The net effect is calcium overload. Mitochondria can mitigate the effects by taking up calcium, but eventually this causes mitochondrial damage and so worsens the energy deficit. As ATP falls further, operation of the sodium pump declines, the ionic gradient is dispersed and the cell depolarizes, causing calcium entry through L-type channels, which worsens the calcium overload and leads to cell death. Positive inotropes, by increasing both cytosolic calcium and energy requirements, will clearly worsen the situation.

This sequence of events has led to development of agents that act via increasing the binding of troponin-C to calcium, rather than by increasing calcium. However, therapy for chronic heart failure is still currently focused on limiting disease progression, largely by interrupting neurohumoral compensation mechanisms. Paradoxically, this includes β-adrenergic receptor antagonists, and also antagonists to angiotensin II and aldosterone (reviewed in Francis (2001)), which may inhibit cardiac remodelling (see below).

Both increased mechanical stress and circulating or local mediators can initiate the gene transcription and protein synthesis associated with cardiac hypertrophy and remodelling. At least part of the response is mediated through calcium-dependent mechanisms involving calcineurin, and isoforms of protein kinase C and calcium–calmodulin-dependent kinase, but growth factors, inflammatory cytokines and other humoral mediators also activate complex signalling cascades that modify activity of transcription factors. Notably, physiological and pathological cardiac remodelling show significant differences in gene transcription and expression of certain proteins; pathological hypertrophy is associated with phenotypic switching towards fetal and less efficient isoforms of contractile proteins such as myosin heavy chain (Hill and Olson 2008).

Chronic heart failure is regarded as irreversible and progressive, and pathological cardiac remodelling plays a major role in its progression. Therapy is largely focused on agents that suppress pro-remodelling pathways (e.g. β-blockers, angiotensin converting enzyme inhibitors, aldosterone antagonists) and therefore slow progression (Francis 2001). However, recent developments have raised hope for future treatments that might reverse cardiac damage.

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